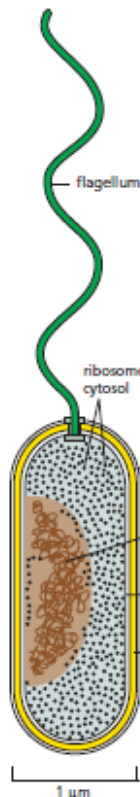
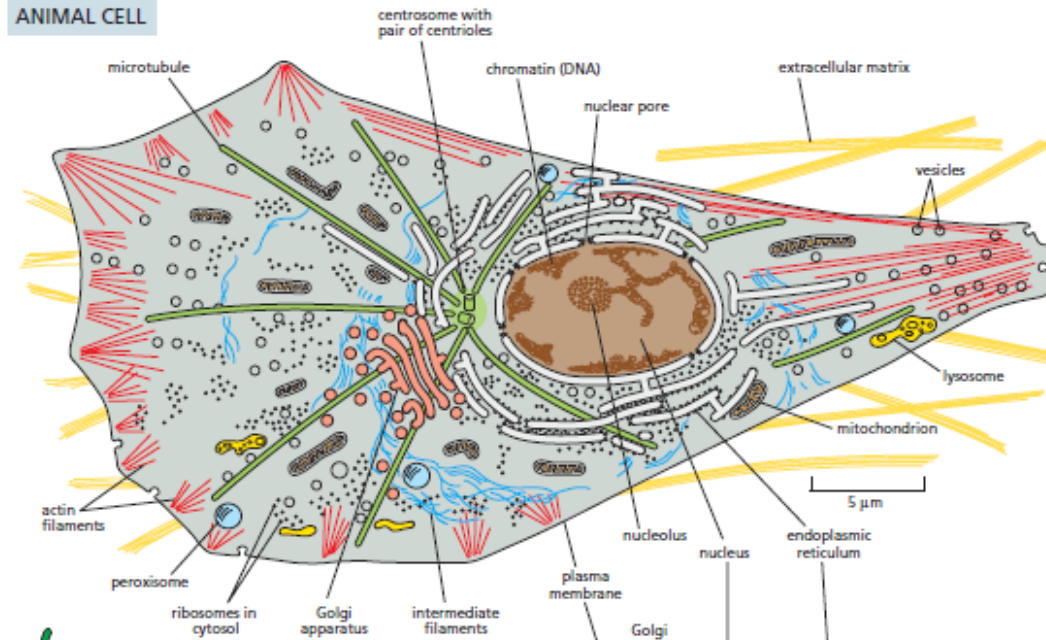


A Tour of the Cell

Unit 1; Chapter 4



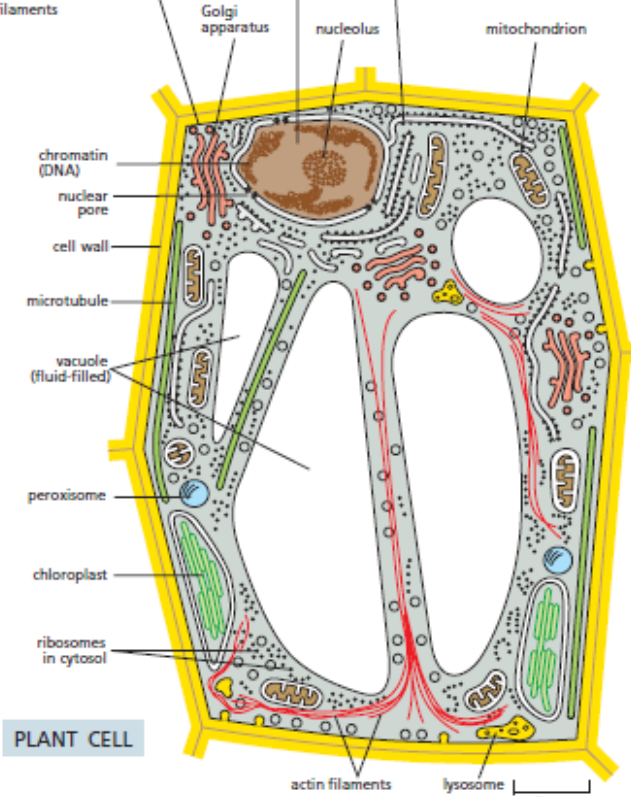
ANIMAL CELL



Three cell types are drawn here in a more realistic manner than in the schematic drawing in Figure 1-24. The animal cell drawing is based on a fibroblast, a cell that inhabits connective tissue and deposits extracellular matrix. A micrograph of a living fibroblast is shown in Figure 1-7A. The plant cell drawing is typical of a young leaf cell. The bacterium shown is rod-shaped and has a single flagellum for motility. A comparison of the scale bars reveals the bacterium's relatively small size.

BACTERIAL CELL

PLANT CELL



Learning Objectives 1

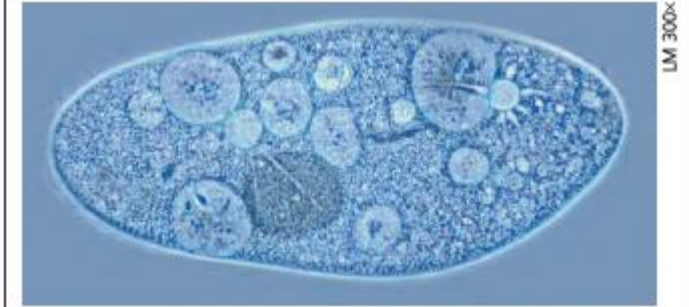
- Compare the following pairs of terms, noting similarities and differences:
 - prokaryotic cells versus eukaryotic cells;
 - light versus electron micrographs.
- Describe the structure of cell membranes and explain why the plasma membrane is called a “fluid mosaic.”
- Explain how antibiotics specifically target bacteria while minimally harming the human host.

The Microscopic World of Cells

- Cell Theory
- Organisms are either
 - Single-celled, such as most prokaryotes (原核生物) and protists (原生生物)
 - Multi-celled
 - plants
 - animals
 - most fungi (真菌)
- Micrographs by microscopes
 - Light, magnify 1,000-fold, live
 - Electron, 100 times more, fix

Light Micrograph (LM)

(can be a view of a living or dead specimen)



Light micrograph of a protist, *Paramecium*

Scanning Electron Micrograph (SEM)

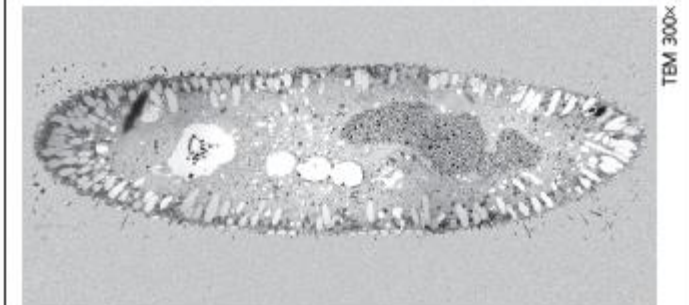
(a view of surface features; specimens must be killed)



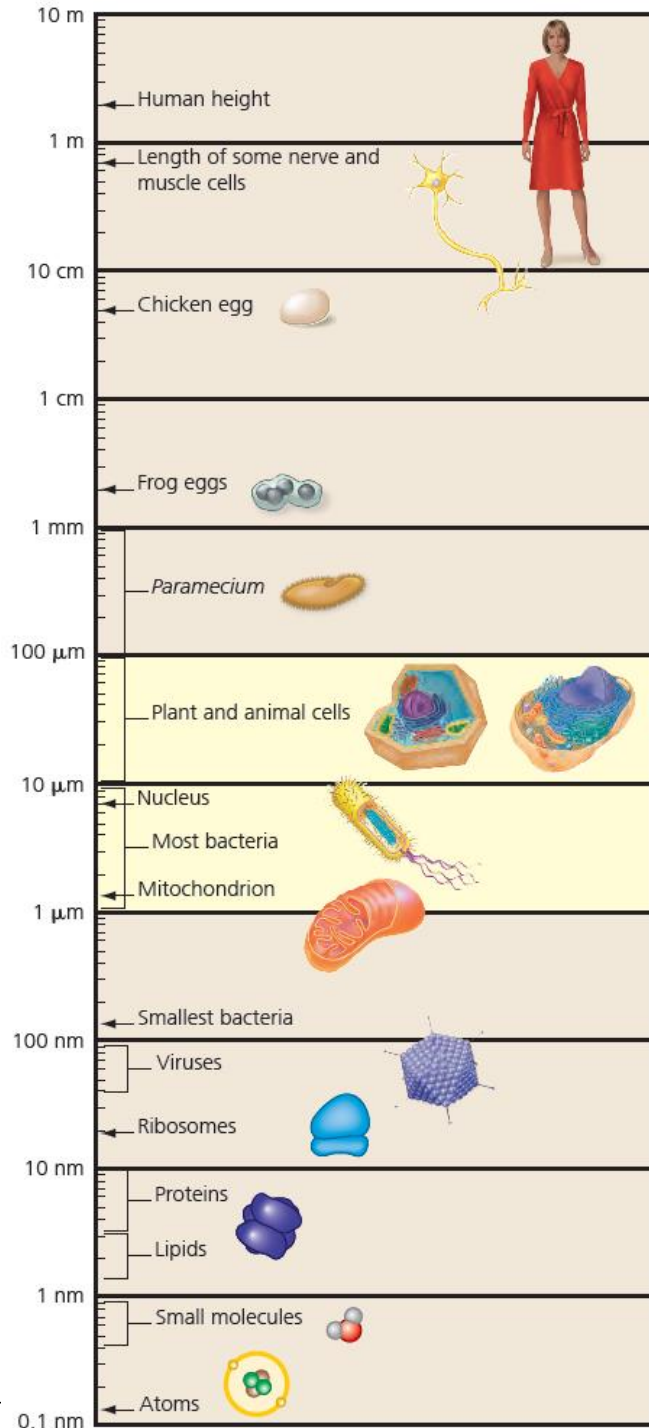
Scanning electron micrograph of *Paramecium*

Transmission Electron Micrograph (TEM)

(a view of internal structures; specimens must be killed)



Transmission electron micrograph of *Paramecium*

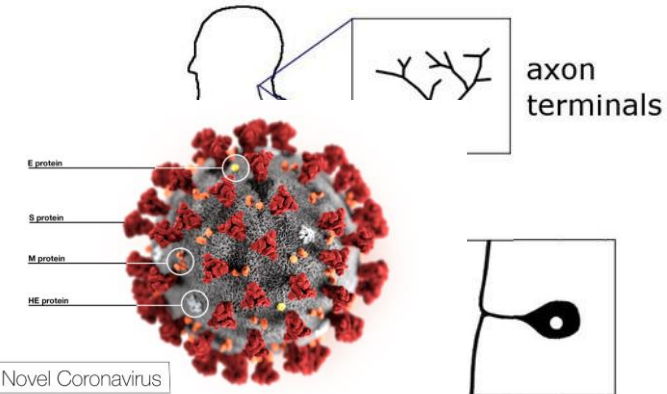


Visible with the naked eye

Light microscope

Electron microscope

The size range of cells compared with objects both larger and smaller.



The SARS-CoV-2 is a large sized virus (approximately 120 nm in diameter).

Measles

receptor terminals

1 meter

1 cent

$$1 \text{ millimeter (mm)} = 10^{-3} \left(\frac{1}{1,000} \right) \text{ m} = \frac{1}{10} \text{ cm}$$

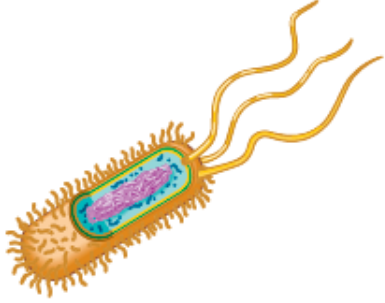
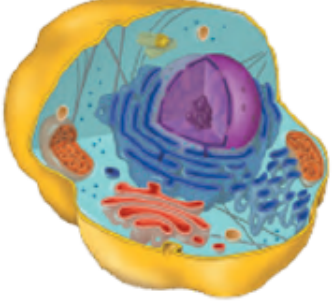
$$1 \text{ micrometer (}\mu\text{m)} = 10^{-6} \text{ m} = 10^{-3} \text{ mm}$$

$$1 \text{ nanometer (nm)} = 10^{-9} \text{ m} = 10^{-3} \mu\text{m}$$

ies

The Two Major Categories of Cells

- Prokaryotic cells
 - Bacteria
 - Archaea
- Eukaryotic cells
 - Protists
 - Plants
 - Fungi
 - Animals

Table 4.1 Comparing Prokaryotic and Eukaryotic Cells	
Prokaryotic cells	Eukaryotic cells
	
First evolved approximately 3.5 billion years ago	First evolved approximately 2.1 billion years ago
Found in bacteria and archaea	Found in protists, plants, fungi, and animals
Smaller, simpler	Larger, more complex
Most have cell walls; some have capsules, fimbriae, and/or flagella.	Plant cells have cell walls; animal cells are surrounded by an extracellular matrix.
Have a plasma membrane	Have a plasma membrane
No membrane-bound organelles	Membrane-bound organelles (for example, nucleus, ER)
Have a nucleoid region containing a single circular chromosome	Have a nucleus containing one or more linear chromosomes
Have ribosomes	Have ribosomes

Several Basic Features For All Cells

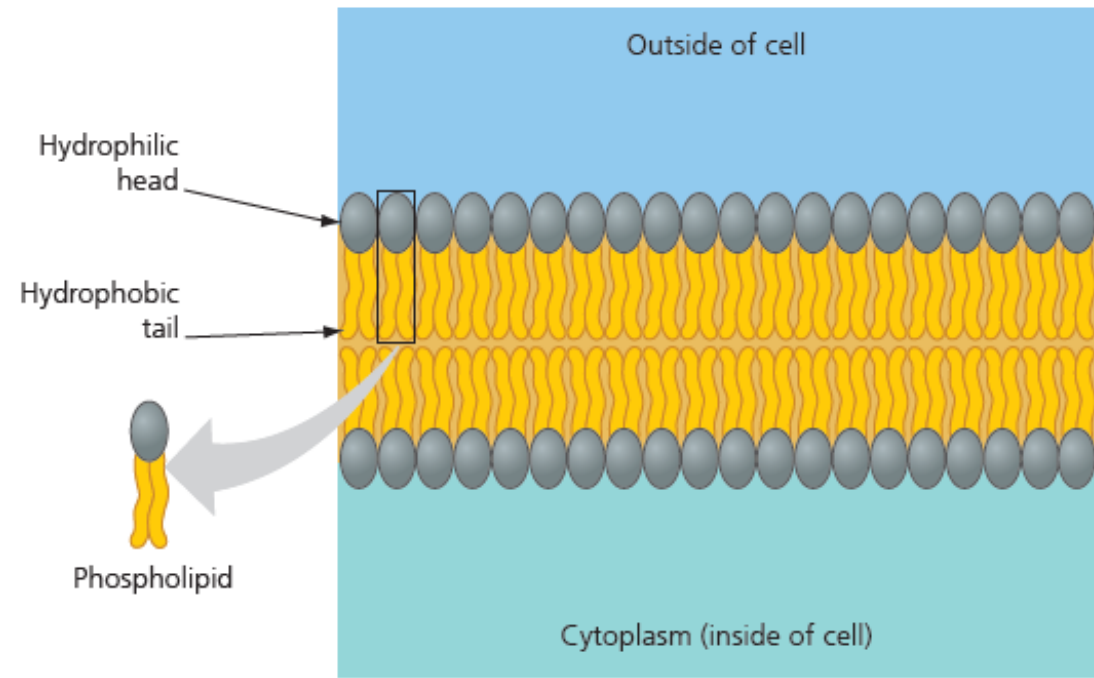
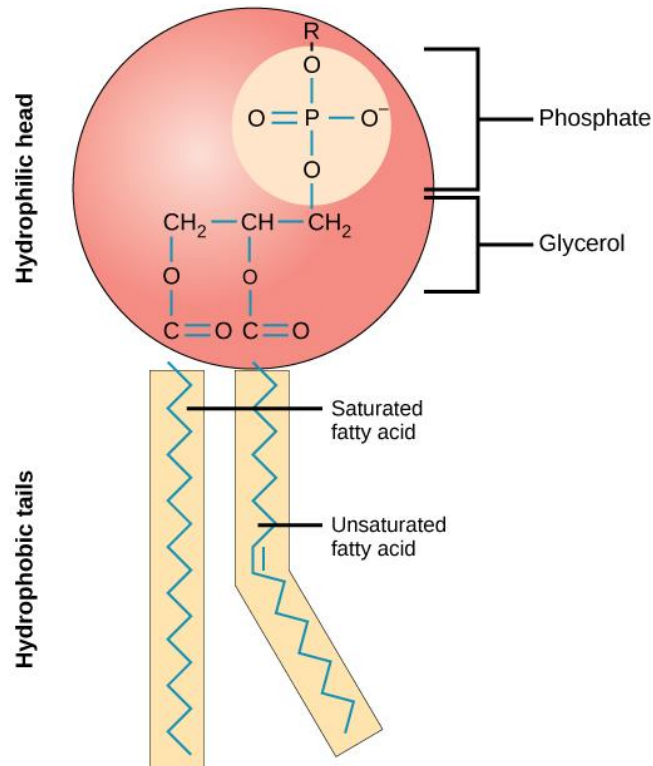
- They are all bounded by a thin **plasma membrane** (细胞质膜).
- Inside all cells is a thick, jelly-like fluid called the **cytosol**, in which cellular components are suspended (squeezed together).
- All cells have one or more **chromosomes** (染色体) carrying genes made of DNA.
- All cells have **ribosomes**, tiny structures that build proteins according to the instructions from the genes.

Cytoplasmic Dynamics and Crowding

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Plasma Membrane

The **plasma membrane and other membranes** of the cell are composed mostly of phospholipids (磷脂), which group together to form a two-layer sheet called a **phospholipid bilayer**.



(a) **Phospholipid bilayer of membrane.** In water, phospholipids

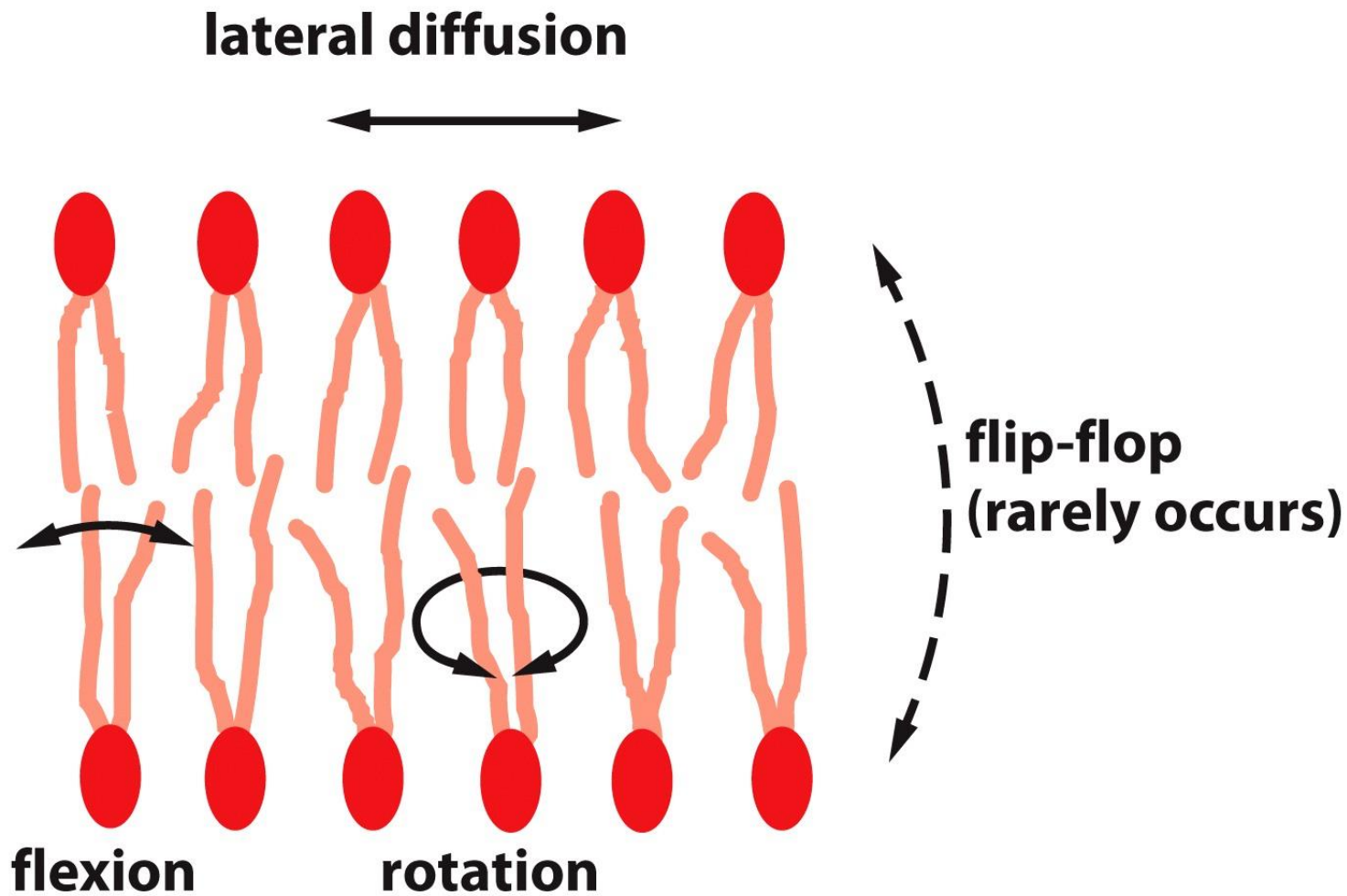


Figure 11-14 Essential Cell Biology, 4th ed. (© Garland Science 2014)

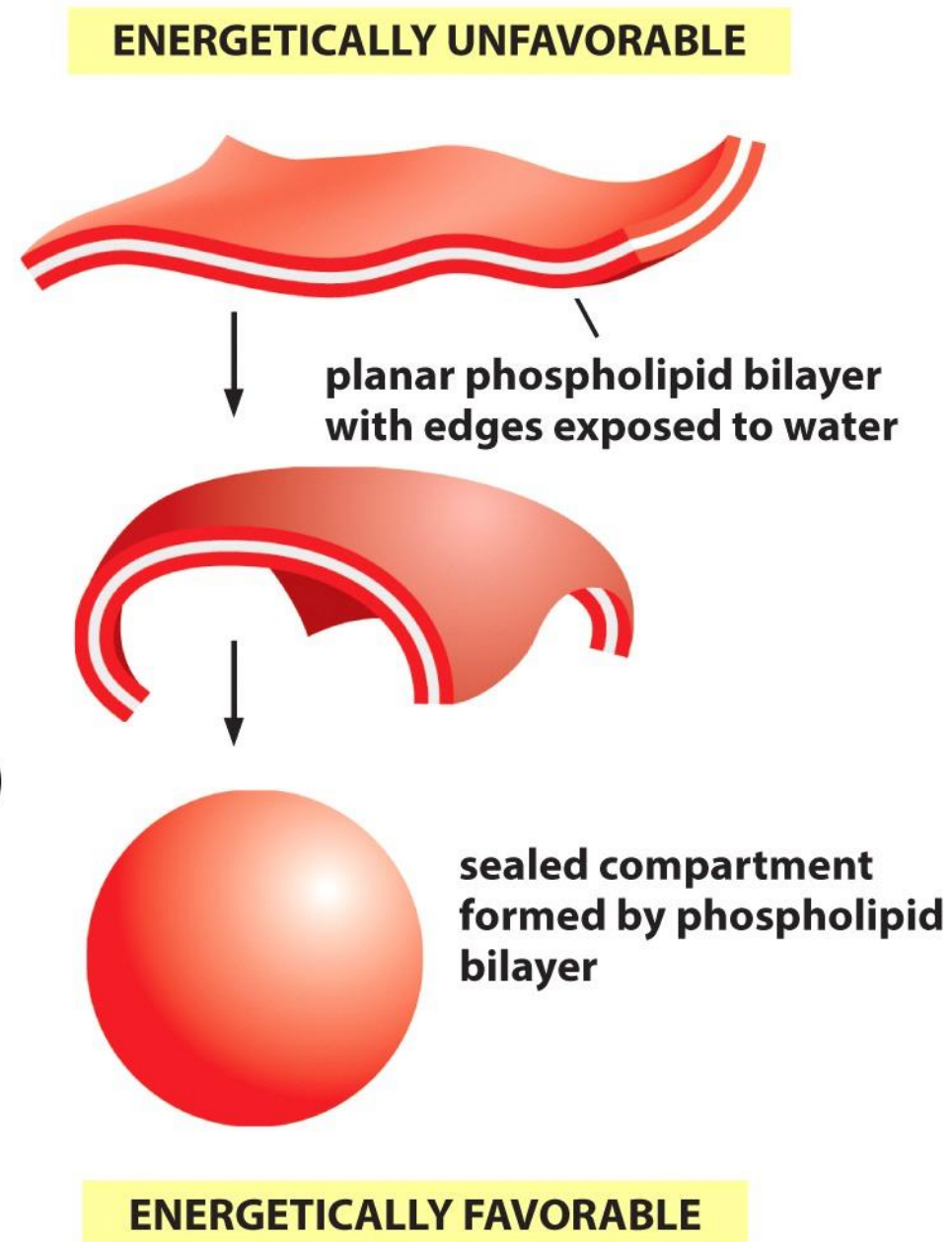
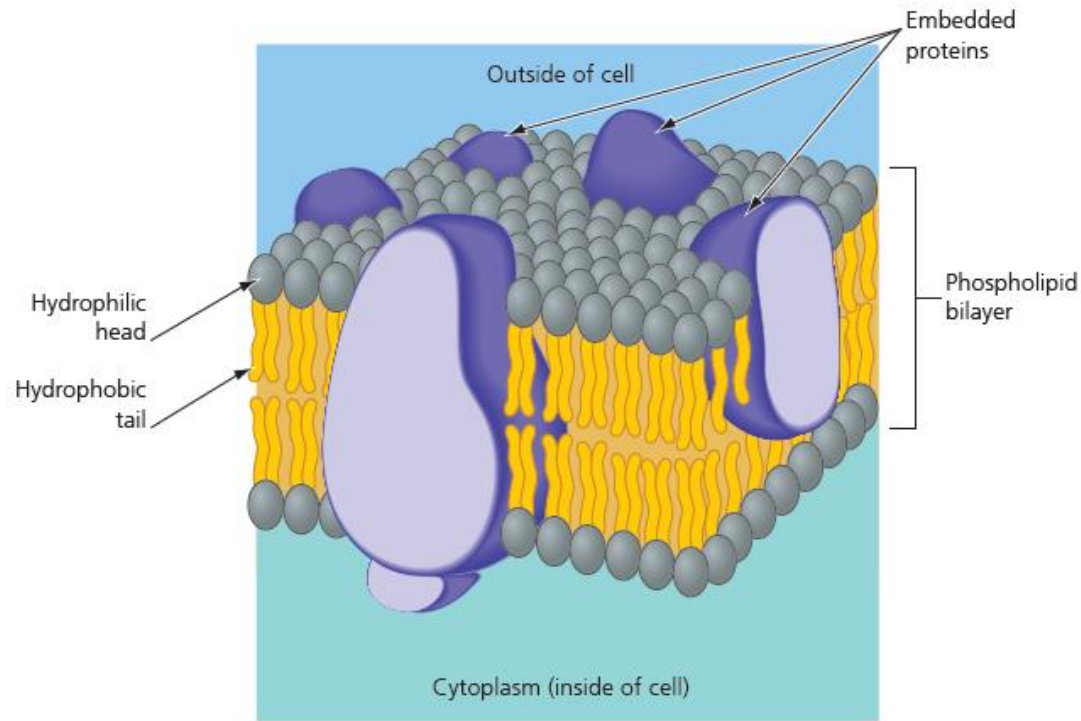


Figure 11-12 Essential Cell Biology, 4th ed. (© Garland Science 2014)

Fluid Mosaic Model of Membrane

- Fluid, molecules can move freely past one another.
- Mosaic (镶嵌), the diversity of proteins *Suspended* in the membrane.



(b) Fluid mosaic model of membrane. Membrane proteins, like the phospholipids, have both hydrophilic and hydrophobic regions.

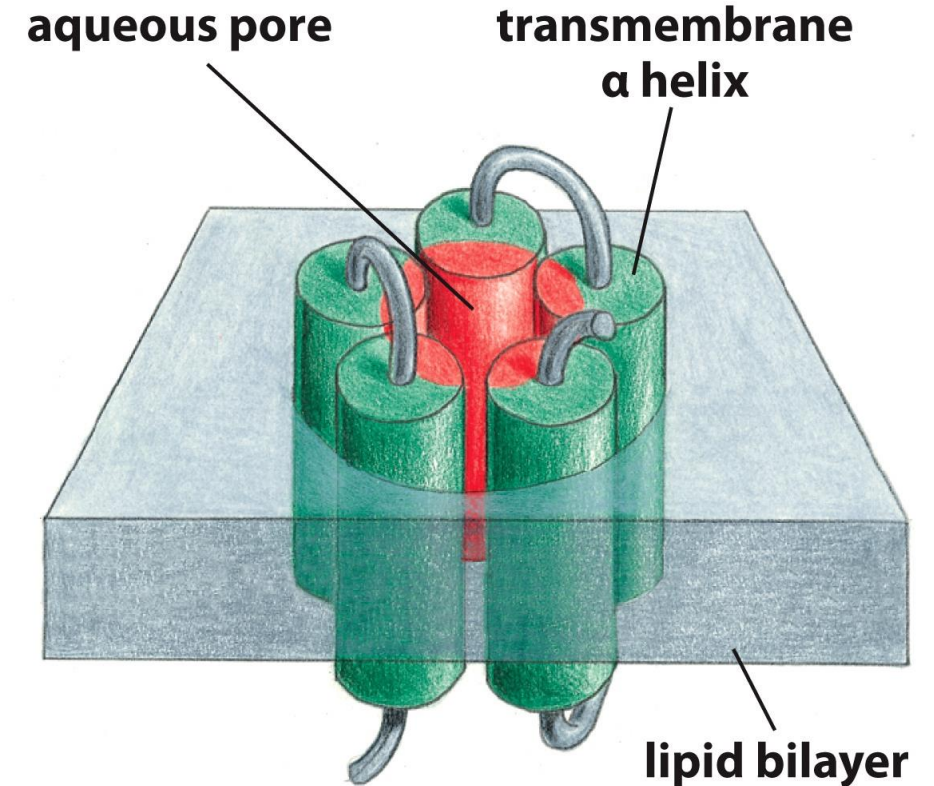


Figure 11-23 Essential Cell Biology, 4th ed. (© Garland Science 2014)

Cholesterol in the Lipid Bilayer

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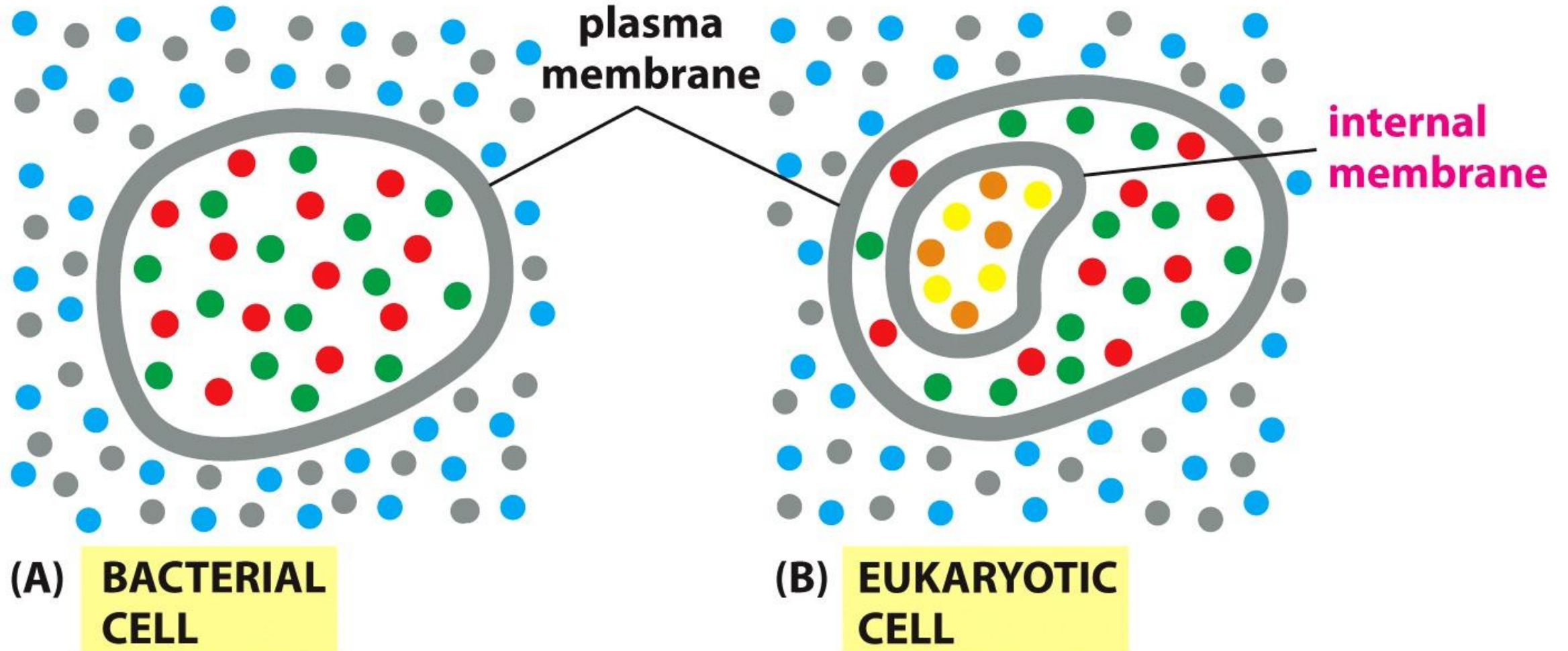


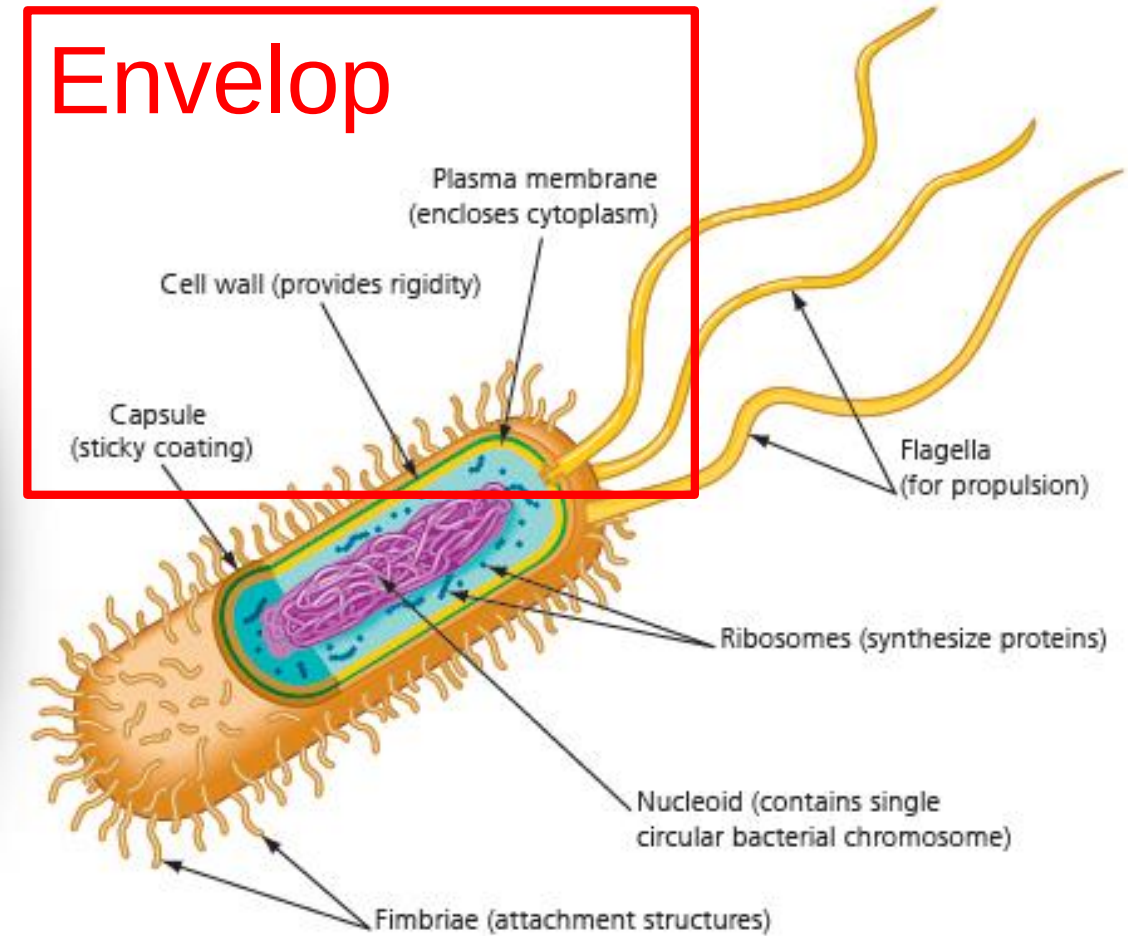
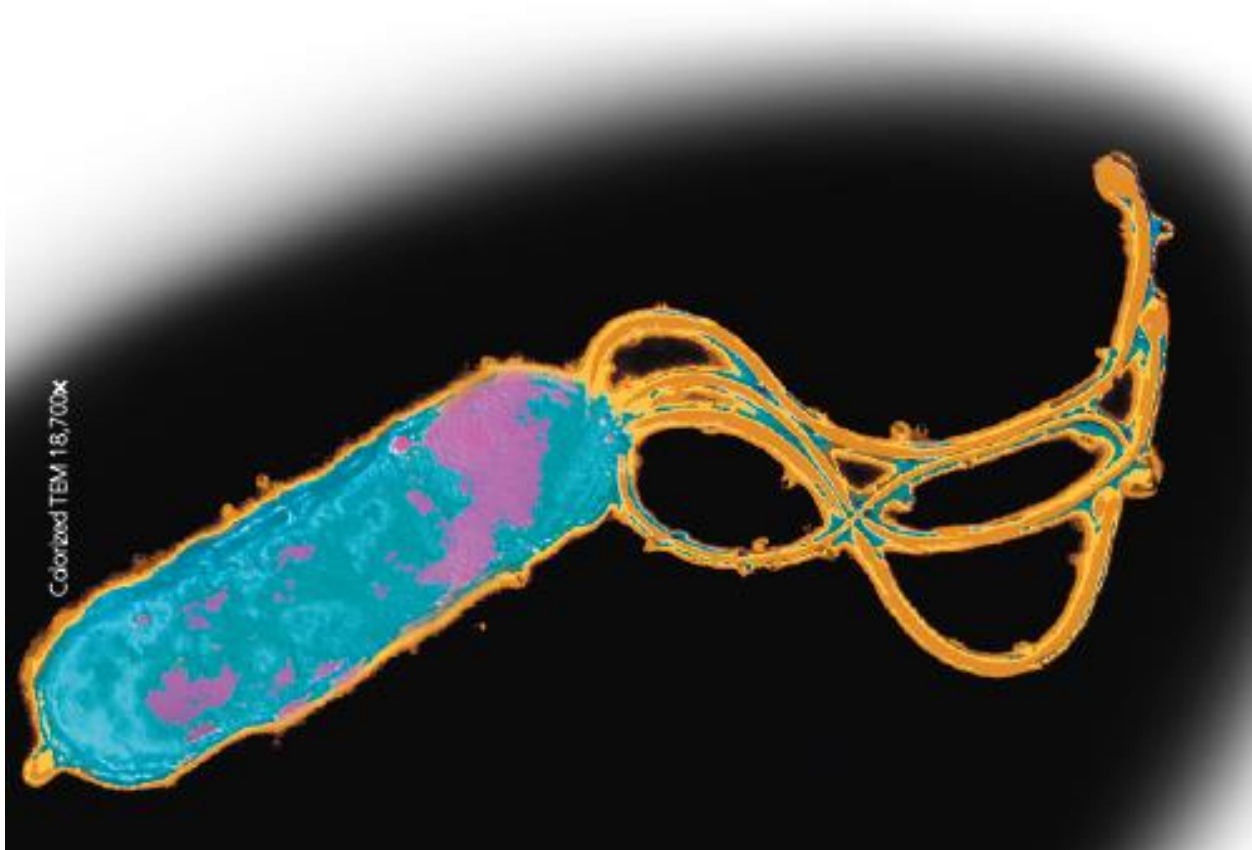
Figure 11-1 Essential Cell Biology, 4th ed. (© Garland Science 2014)

Prokaryotes vs. Eukaryotes

- Prokaryotic cells are **older**.
 - 3.5 billion years vs 2.1 billion years
- Prokaryotic cells are **smaller and simpler**.
- Only eukaryotic cells have membrane-enclosed **organelles** (“little organs”, 细胞器), **membrane-enclosed structures** that perform specific functions.
 - The most prominent organelle is the **nucleus** (细胞核), which houses most (not all) of a eukaryotic cell’s DNA and is surrounded by a double membrane.
 - A prokaryotic cell lacks a nucleus. Its DNA is coiled into a nucleus-like **region** called the **nucleoid** (拟核).

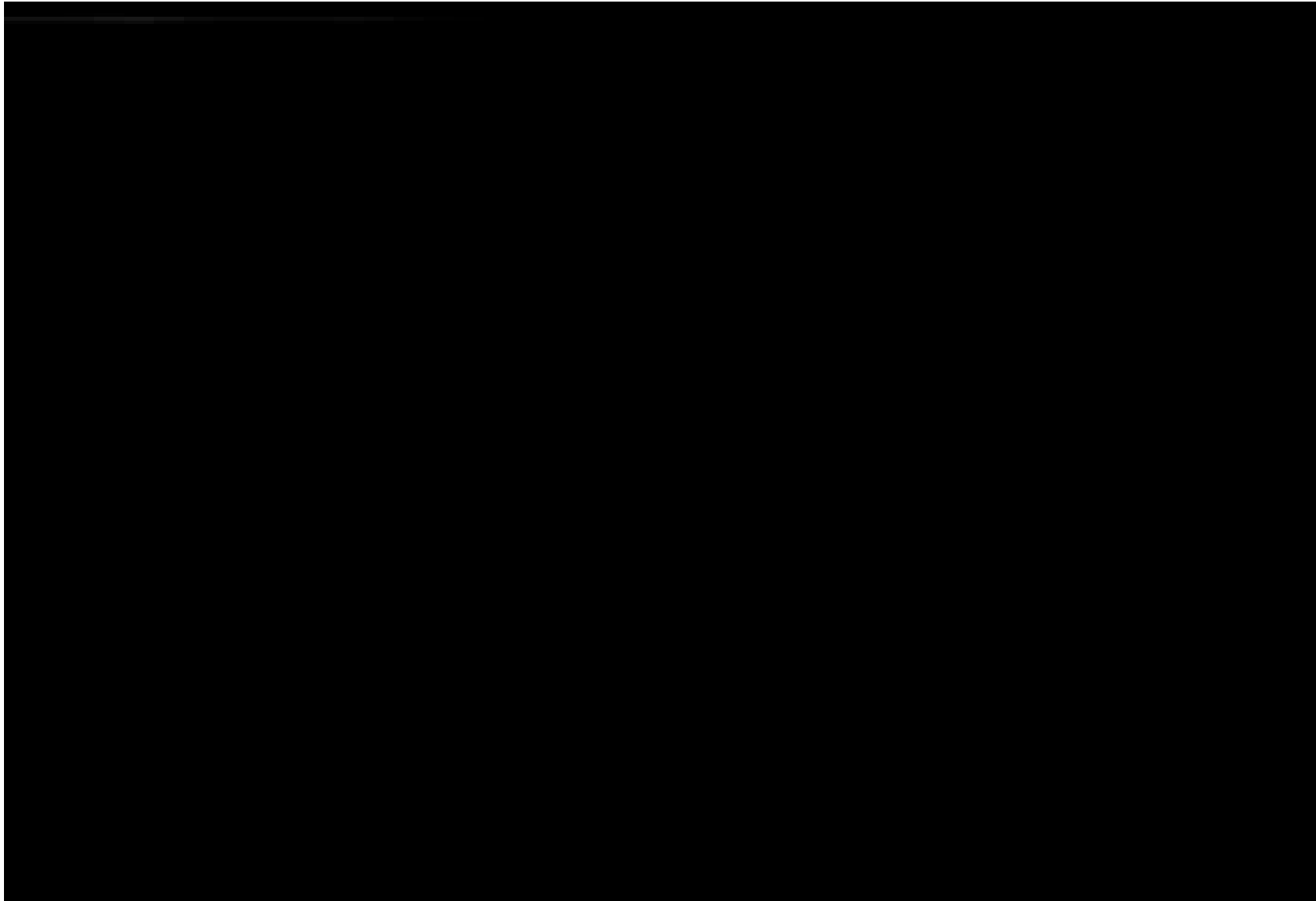
A Prokaryotic Cell

▼ **Figure 4.3** A prokaryotic cell. A micrograph of *Helicobacter pylori* (left), a bacterium that causes stomach ulcers, is shown alongside a drawing of an idealized prokaryotic cell (right).



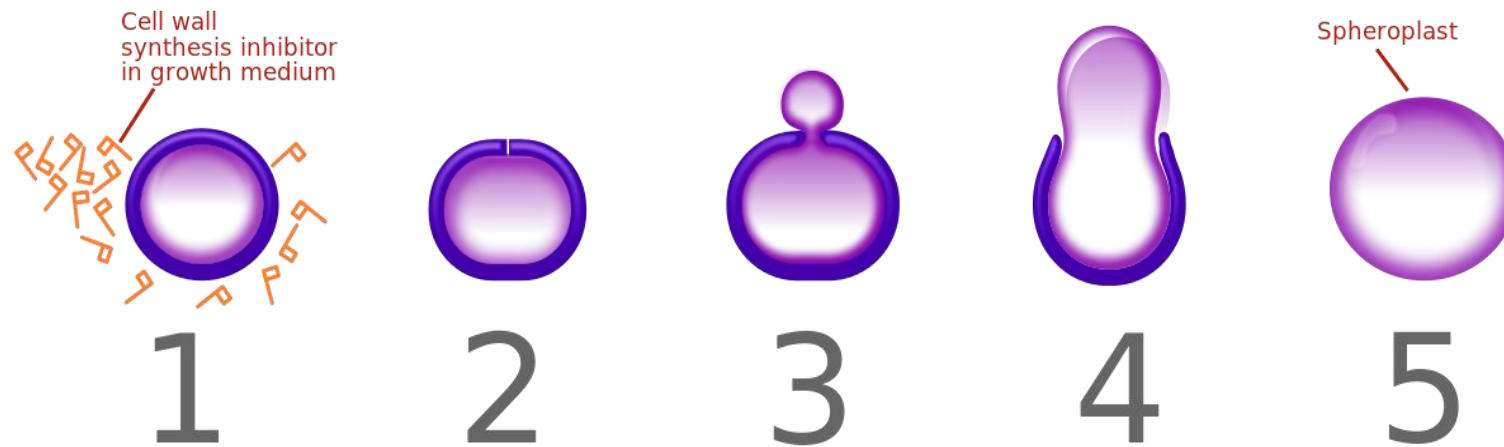
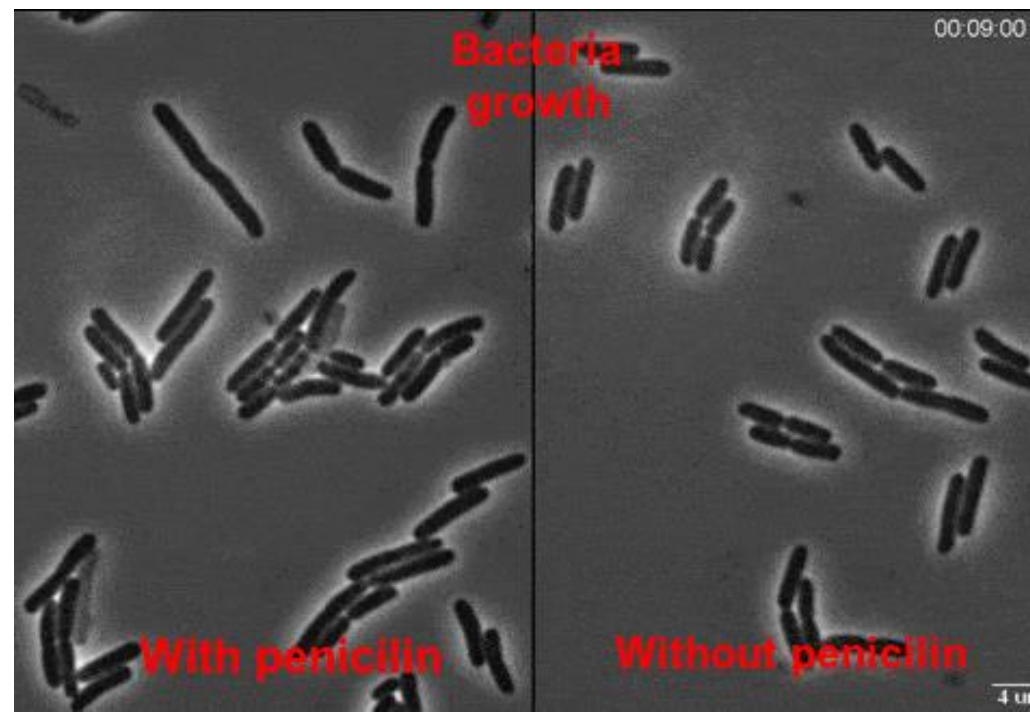
Pili

Prokaryotic Flagella (*Salmonella typhimurium*)



Antibiotics: Drugs That Target Bacterial Cells

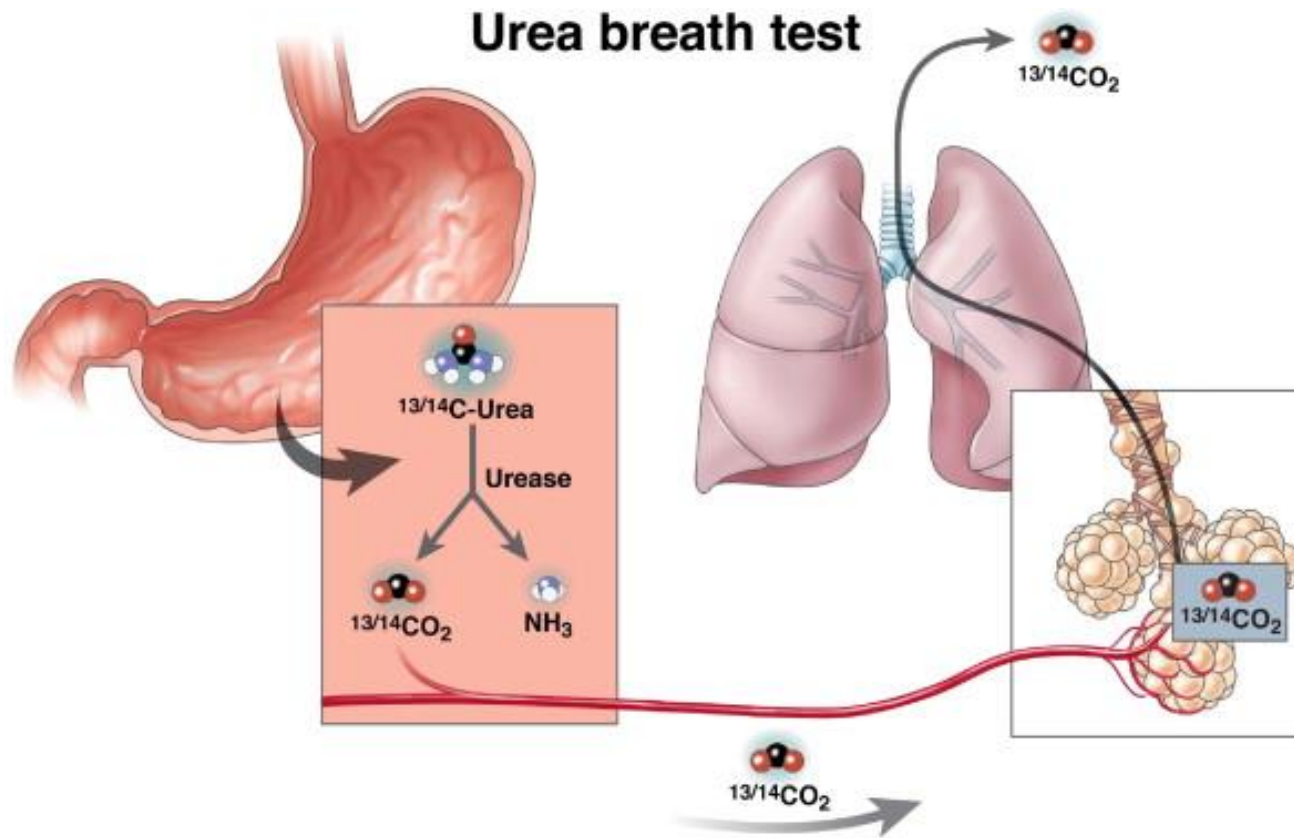
- The first antibiotic (抗生素) to be discovered was penicillin in 1920.
- The widespread use of antibiotics drastically decreased deaths from bacterial infections.
- Most antibiotics bind to structures found **only in bacterial cells**.
 - Some antibiotics bind to the bacterial ribosomes, leaving human ribosomes unaffected.
 - Other antibiotics target enzymes found only in the bacterial cells.



Bacteria that attempt to grow and divide in the presence of penicillin fail to do so, and instead end up shedding their cell walls.

Helicobacter Pylori (幽门螺旋菌)

- Detected by Urea Breath Test



Diagnosis- C13/C14 Breath Test

Nobel Prize in Physiology or Medicine 2005



Photo: C. Northcott
Barry J. Marshall

Prize share: 1/2



Photo: U. Montan
J. Robin Warren

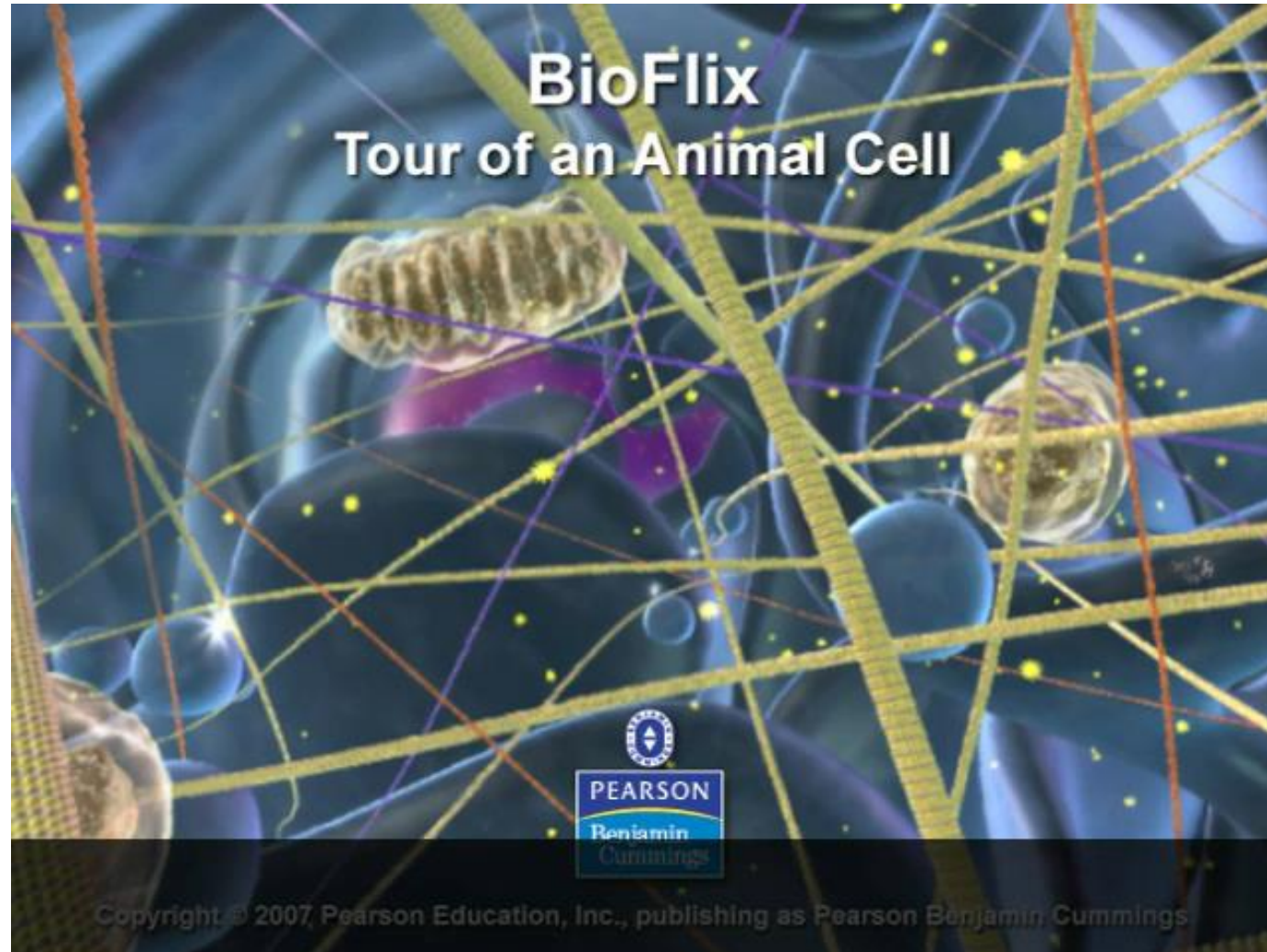
Prize share: 1/2

The Nobel Prize in Physiology or Medicine 2005 was awarded jointly to Barry J. Marshall and J. Robin Warren "for their discovery of the bacterium *Helicobacter pylori* and its role in gastritis and peptic ulcer disease"

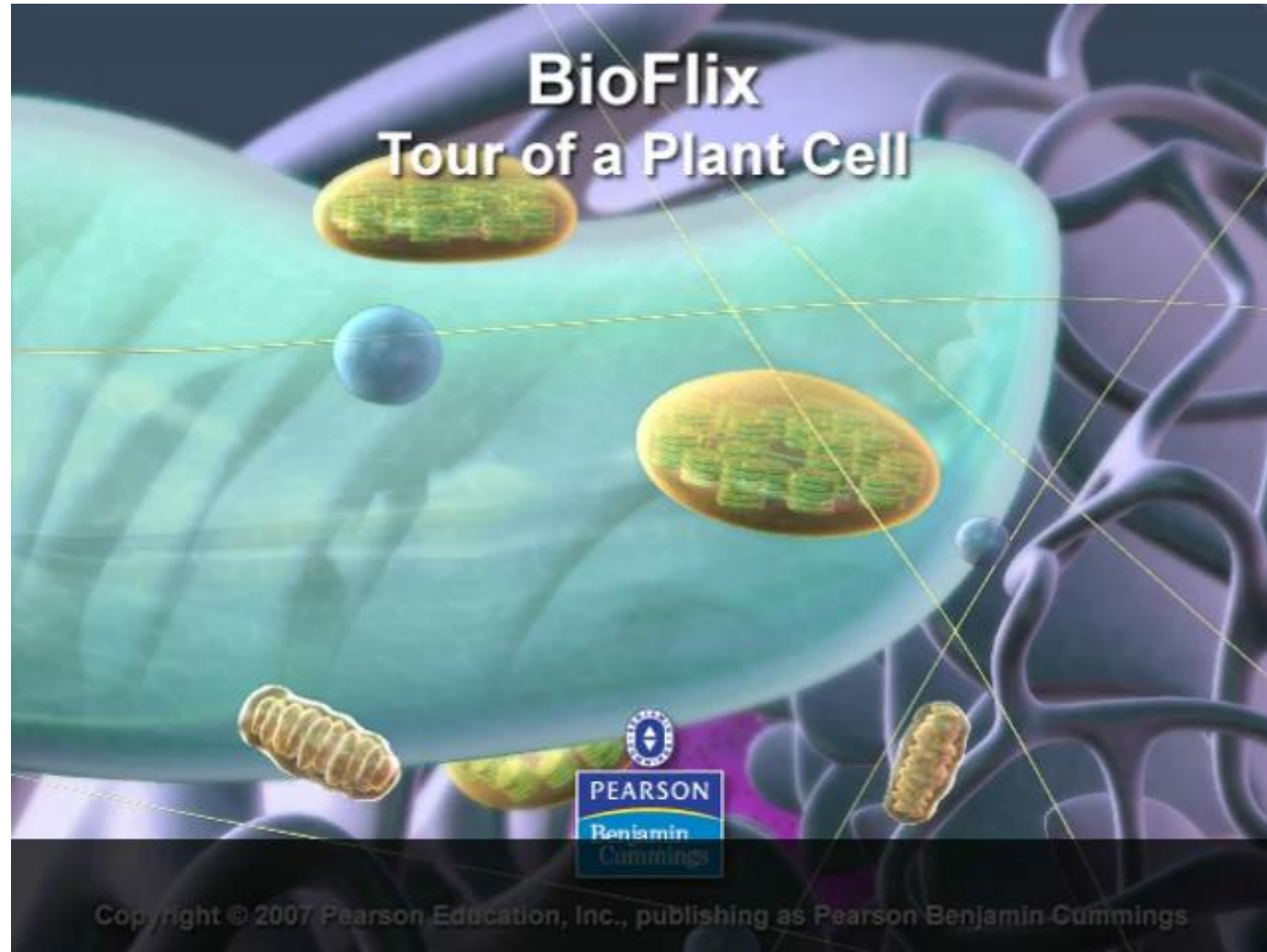
Learning Objectives 2

- Compare the structures and functions of a plant cell wall with the extracellular matrix of an animal cell.
- Explain how the genetic information in the nucleus is used to direct the production of proteins in the cytoplasm.
- Compare the structures and functions of the following components of the endomembrane system: nuclear envelope, rough endoplasmic reticulum, smooth endoplasmic reticulum, Golgi apparatus, lysosomes, and vacuoles.

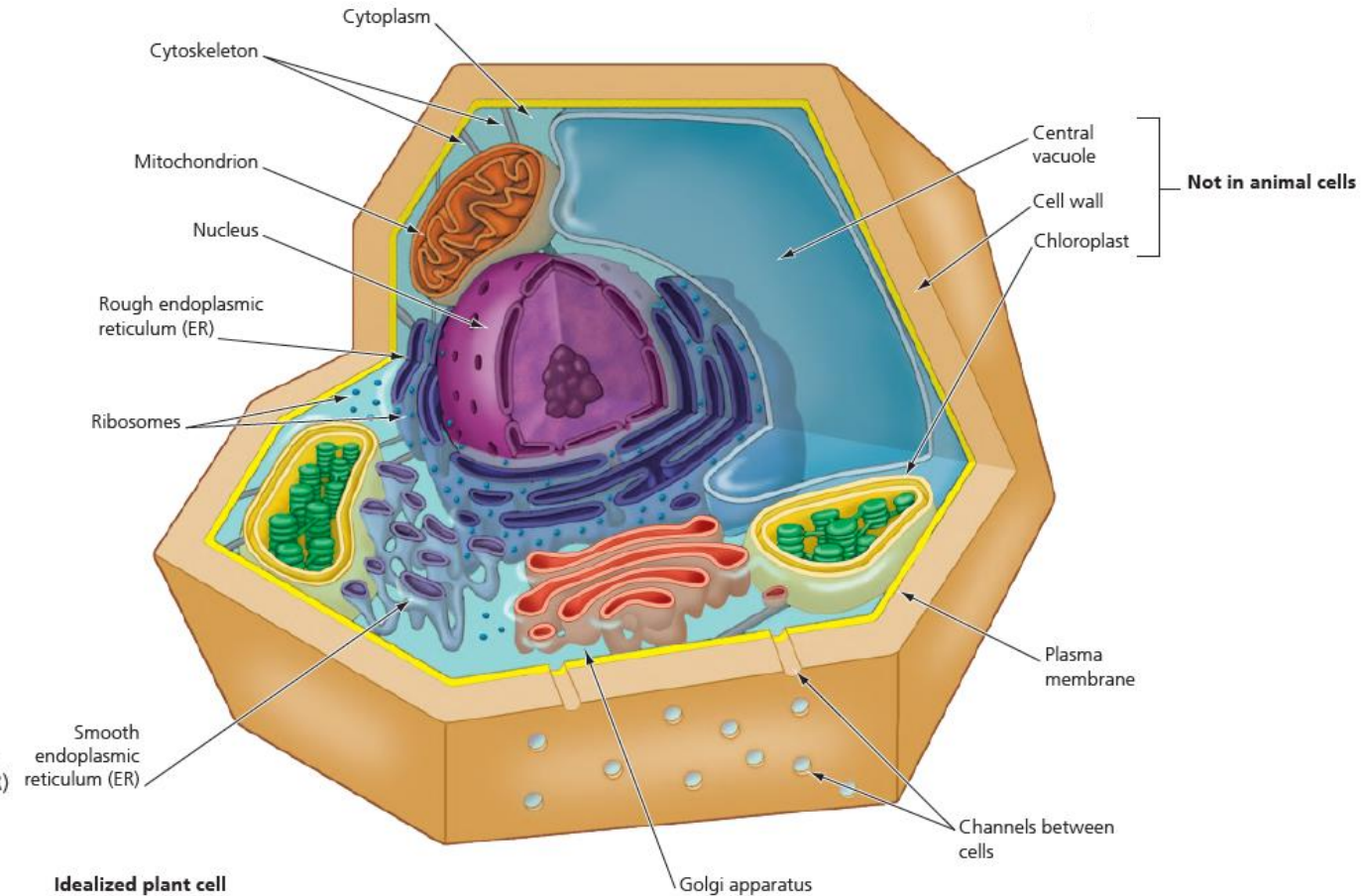
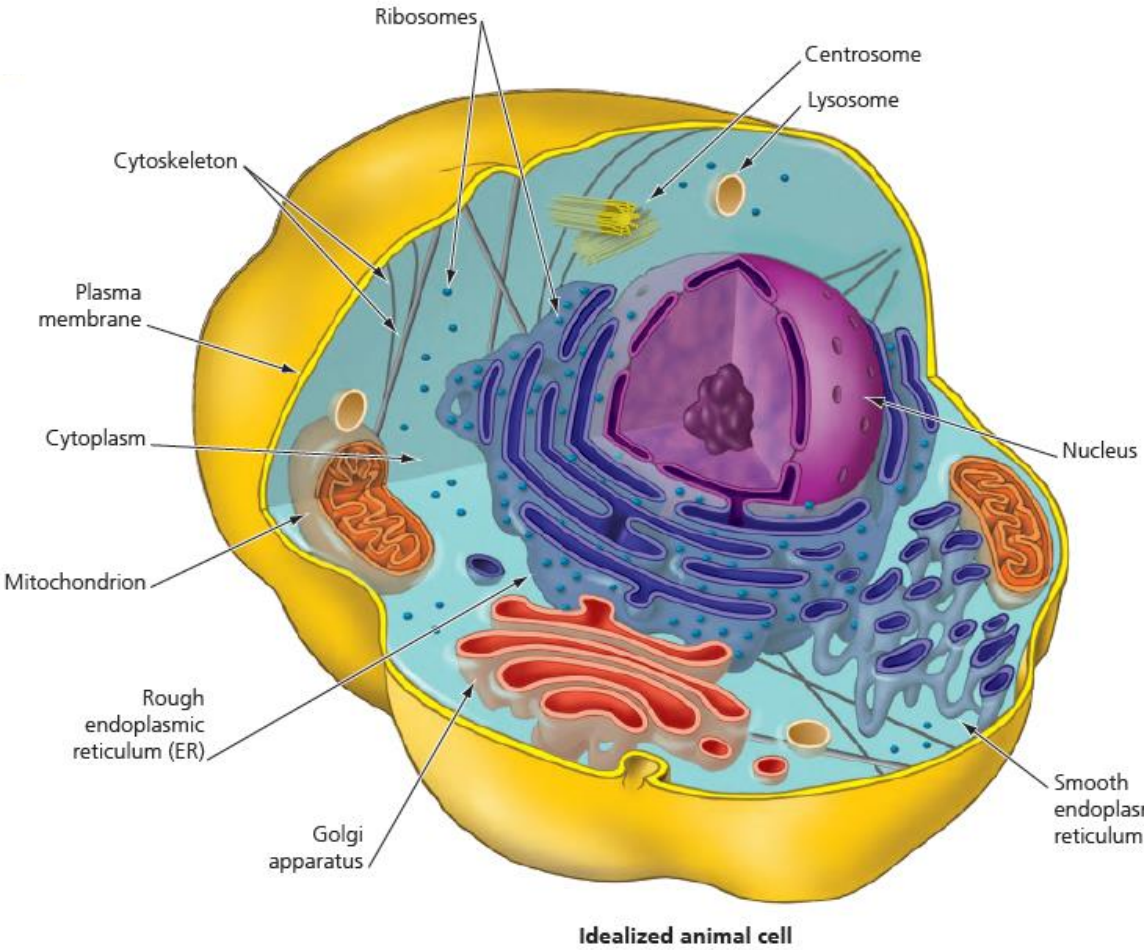
Tour of an Animal Cell



Tour of a Plant Cell



Animal vs. Plant Cells

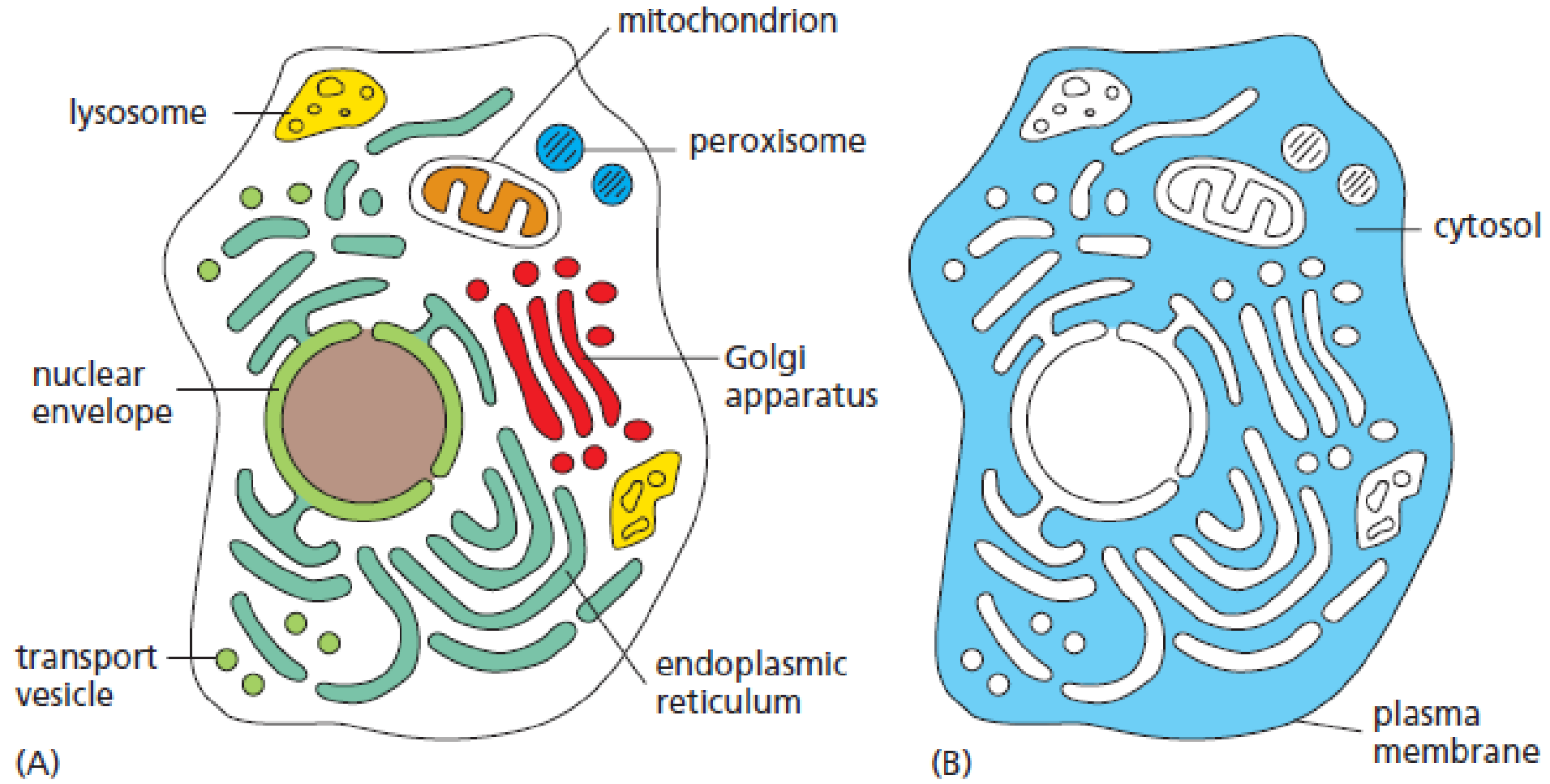


Cell Surfaces

- Plant cells have a **cell wall** made from cellulose fibers.
 - protect the cells
 - maintain cell shape
 - keep cells from absorbing too much water
- Animal cells (lack cell walls) are supported by fibers made of the protein collagen (胶原蛋白)
 - Hold cells together in tissues
 - Protective and supportive functions
 - **Cell junctions** the surfaces of most animal cells
 - connect cells together into tissues
 - coordinate cell function

Cytoplasm: region between the nucleus and plasma membrane

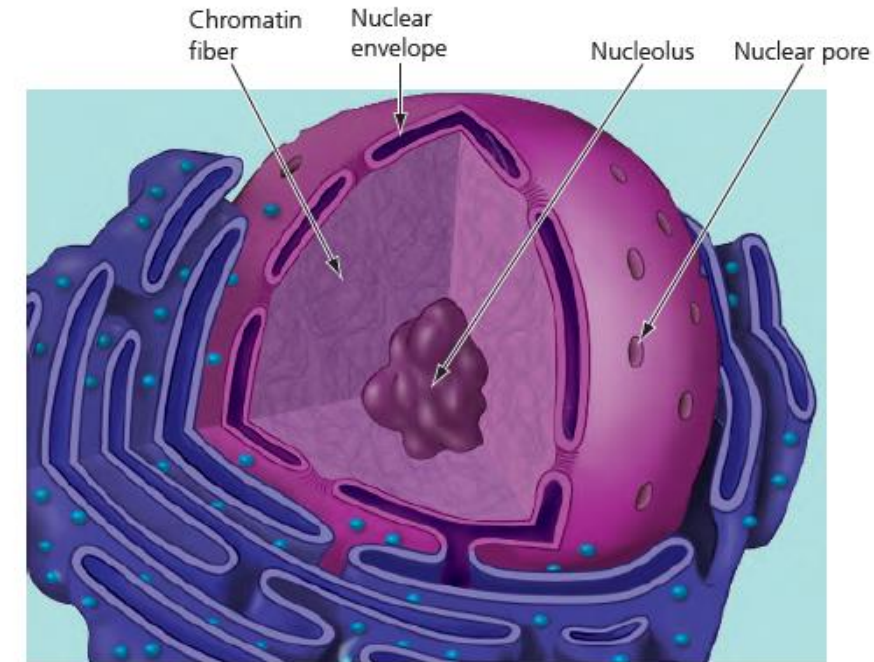
Cytosol+Organelles



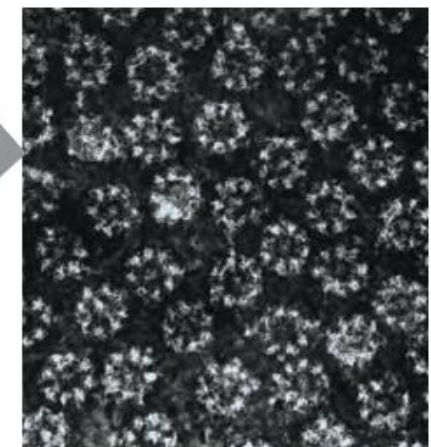
The Nucleus and Ribosomes: Genetic Control of the Cell

- The nucleus is separated from the cytoplasm by a **double membrane** called the **nuclear envelope** (核膜) with **nuclear pores**.
- **Chromatin** (染色质) long **DNA** molecules and associated **proteins** form fibers.
 - Each long chromatin fiber constitutes one chromosome.
- The **nucleolus** (核仁) a prominent structure within the nucleus where the components of **ribosomes** are made.

▼ Figure 4.7 The nucleus.

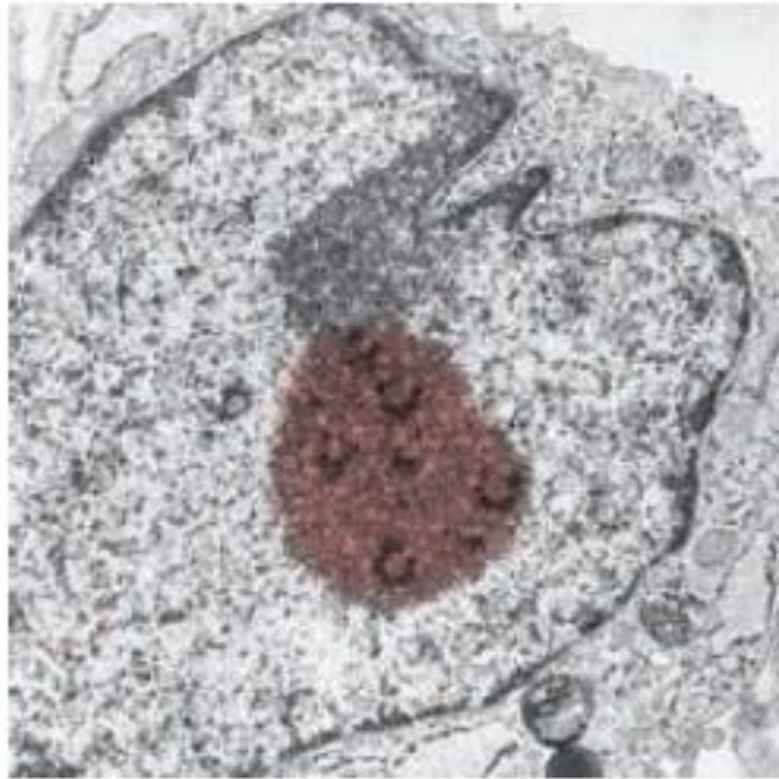


Surface of nuclear envelope



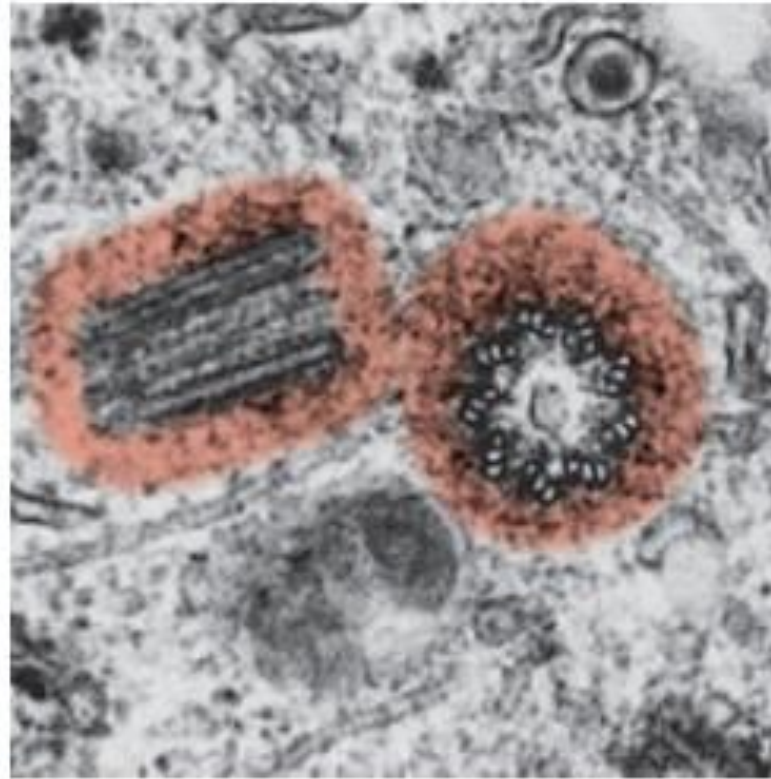
Nuclear pores

Intracellular Condensates Form Biochemical Subcompartments by **Phase Separation**



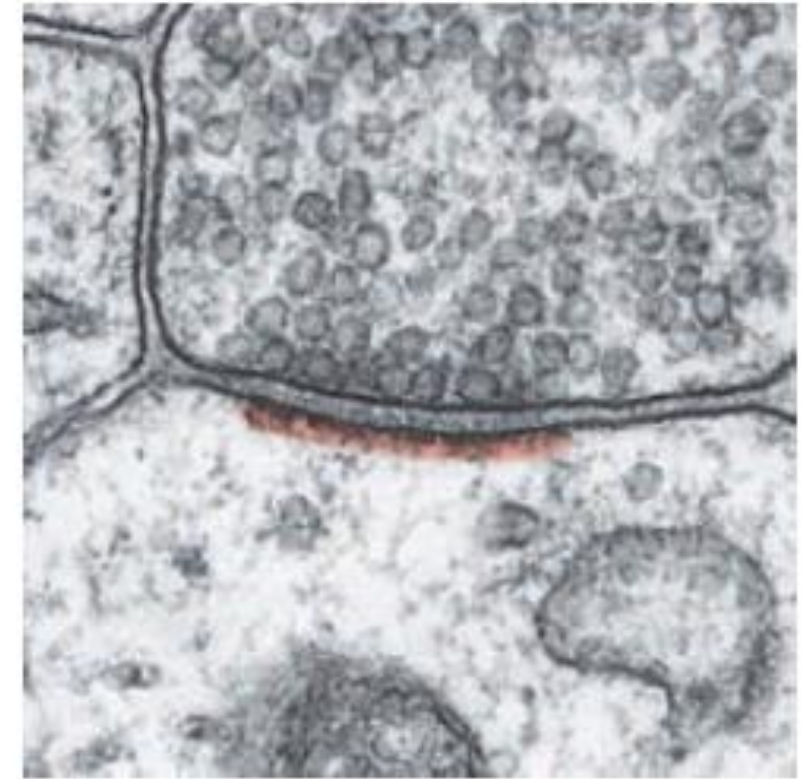
(B)

2 μm



(C)

1 μm



(D)

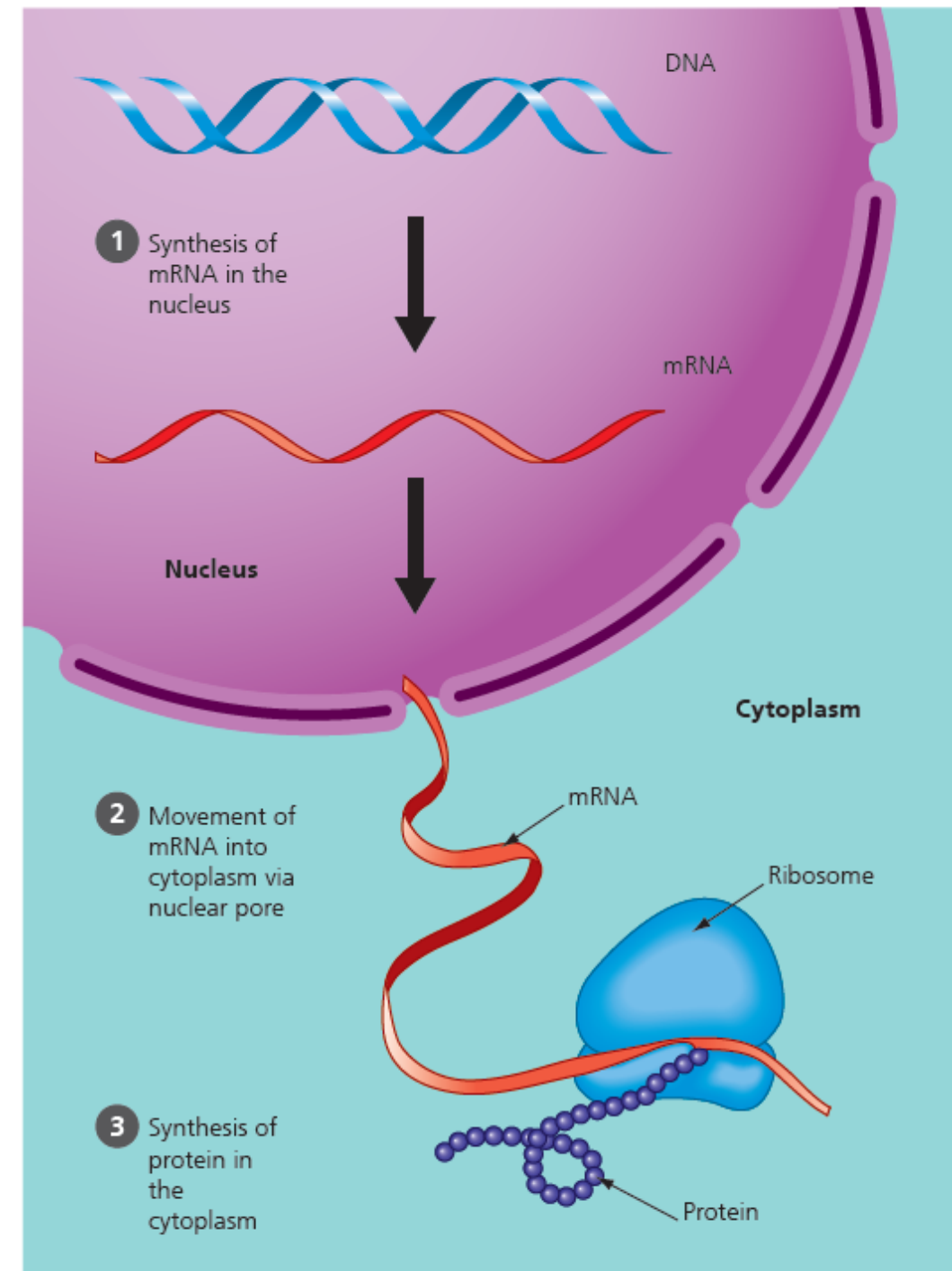
1 μm

The Nucleus and Ribosomes: Genetic Control of the Cell

- Ribosomes are responsible for **protein synthesis**.
- In eukaryotic cells, the components of ribosomes are made in the nucleus and then **transported through the pores** of the nuclear envelope into the cytoplasm, where ribosomes begin their work.
- Structurally identical large and small subunits:
 - Suspended in the cytosol, proteins that **remain** within the cell
 - Attached to the outside of the nucleus or the endoplasmic reticulum, proteins that are **incorporated** into membranes or **secreted** by the cell.

Two Questions

- The adaptive advantage of using mRNA to direct the production of proteins instead of using DNA directly.
- How we can have **four main types** of organic molecules functioning in specific roles in our cells, yet DNA and RNA only specifically dictate the generation of proteins, and more copies of DNA and RNA? How is the production of specific types of carbohydrates and lipids in cells controlled?

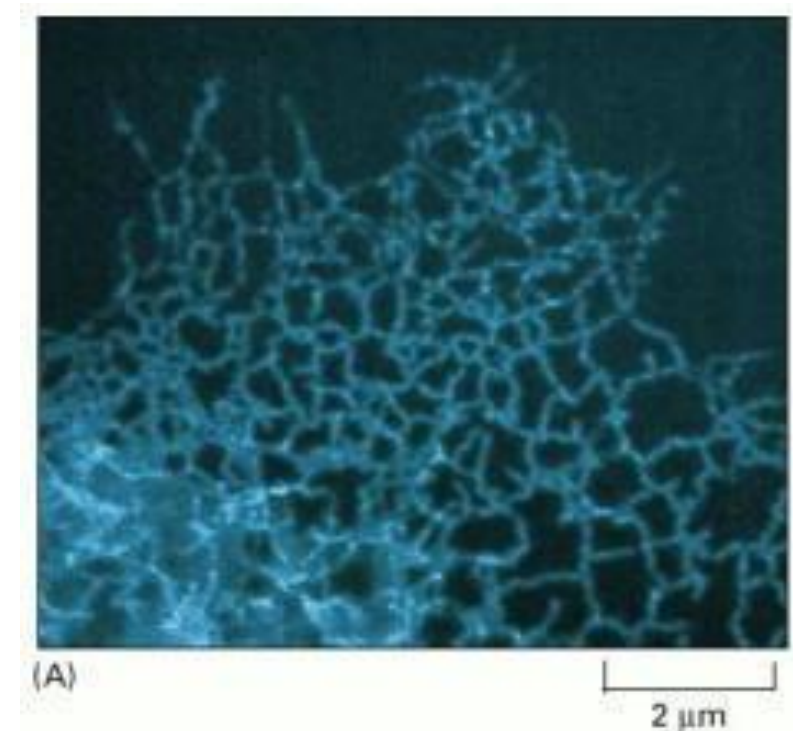
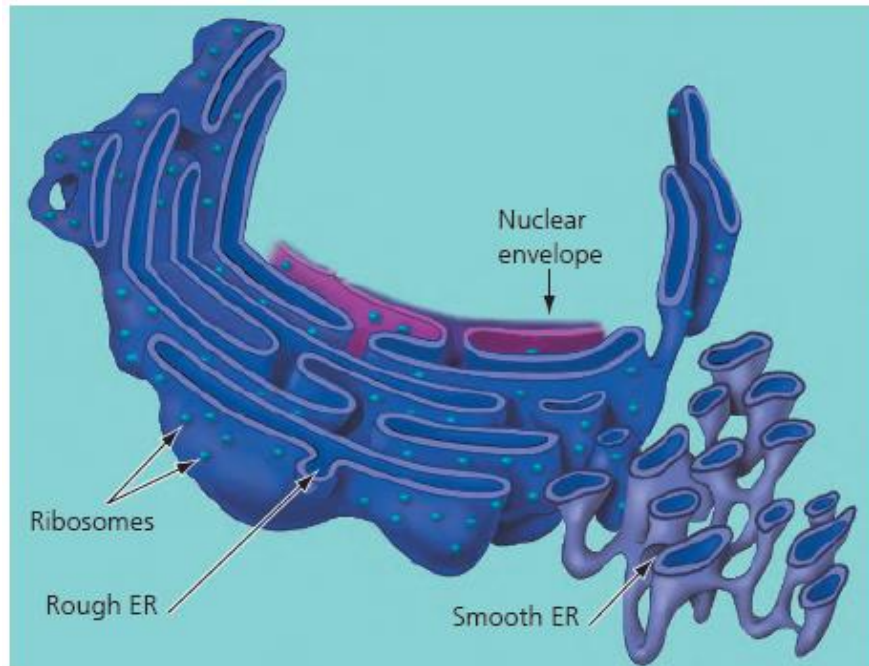


The Endomembrane System: Manufacturing and Distributing Cellular Products

- The endomembrane system (内膜系统) in a cell consists of
 - the nuclear envelope
 - the endoplasmic reticulum
 - the Golgi apparatus
 - lysosomes
 - Vacuoles
- These membranous organelles are either
 - physically connected or
 - linked by vesicles (囊泡), sacs made of membrane.

The Endoplasmic Reticulum

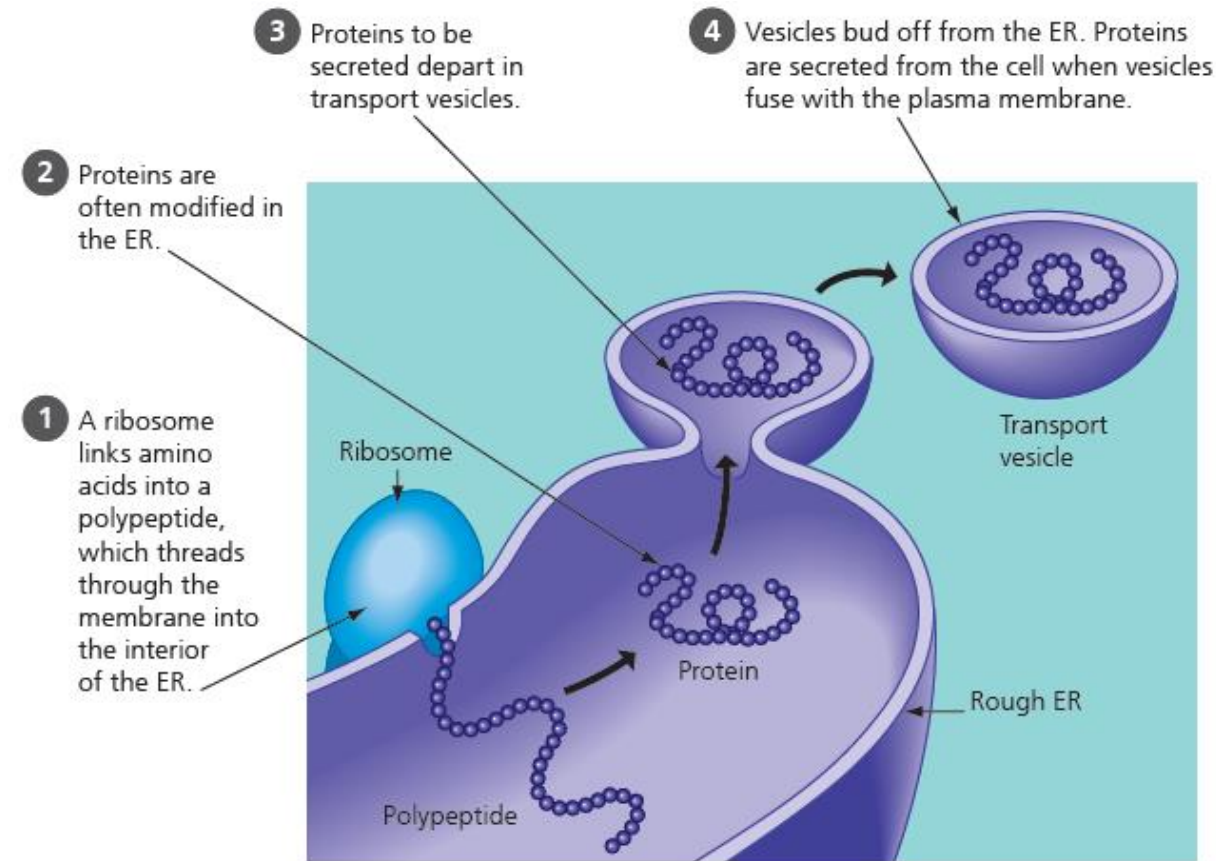
- One of the main manufacturing facilities in a cell:
 - produces an enormous variety of molecules (including membrane)
 - connected to the nuclear envelope
 - composed of interconnected **rough and smooth ER** that have different structures and functions

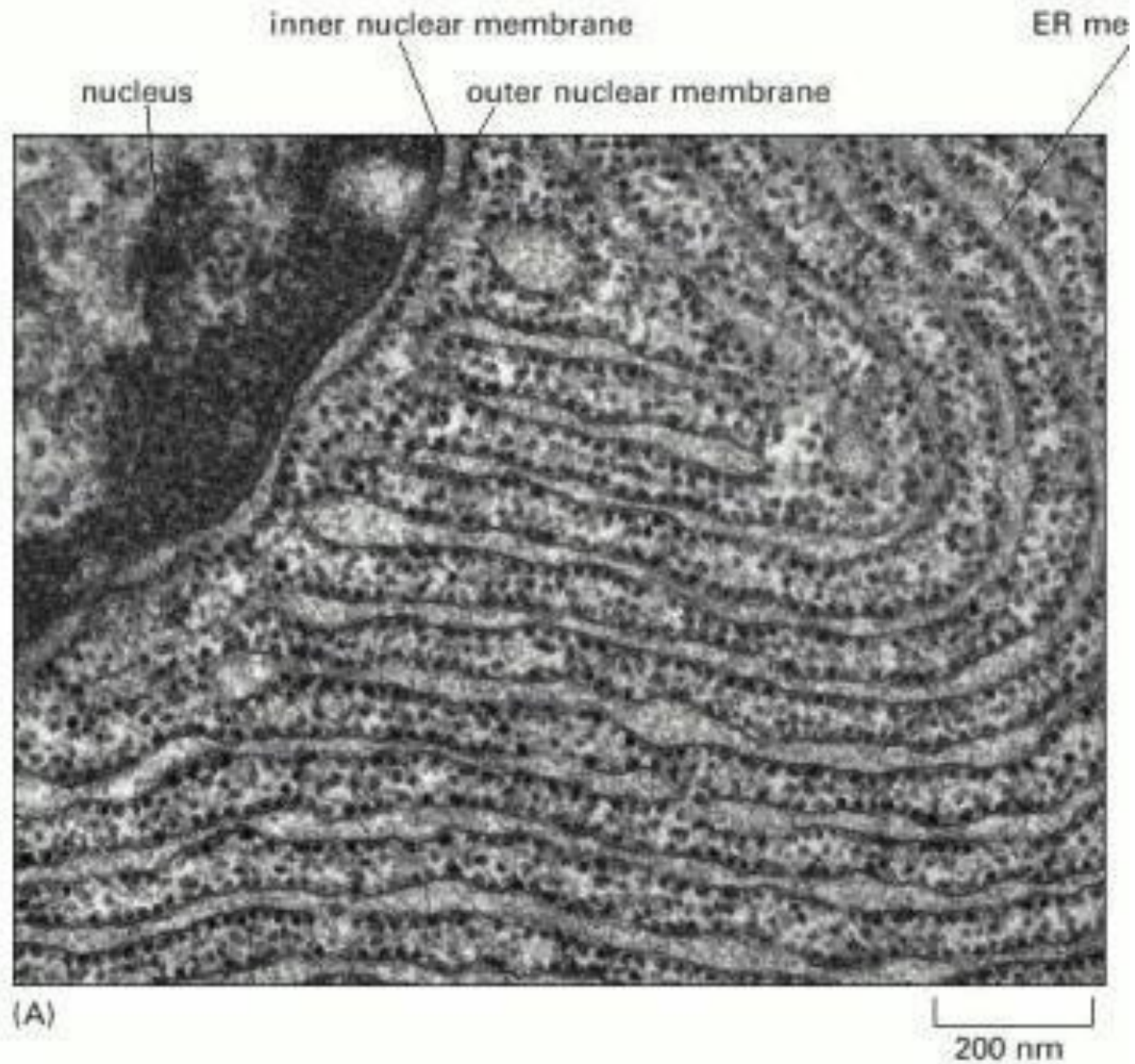


The Endoplasmic Reticulum: Rough ER

The “rough” refers to ribosomes that stud the outside of its membrane.

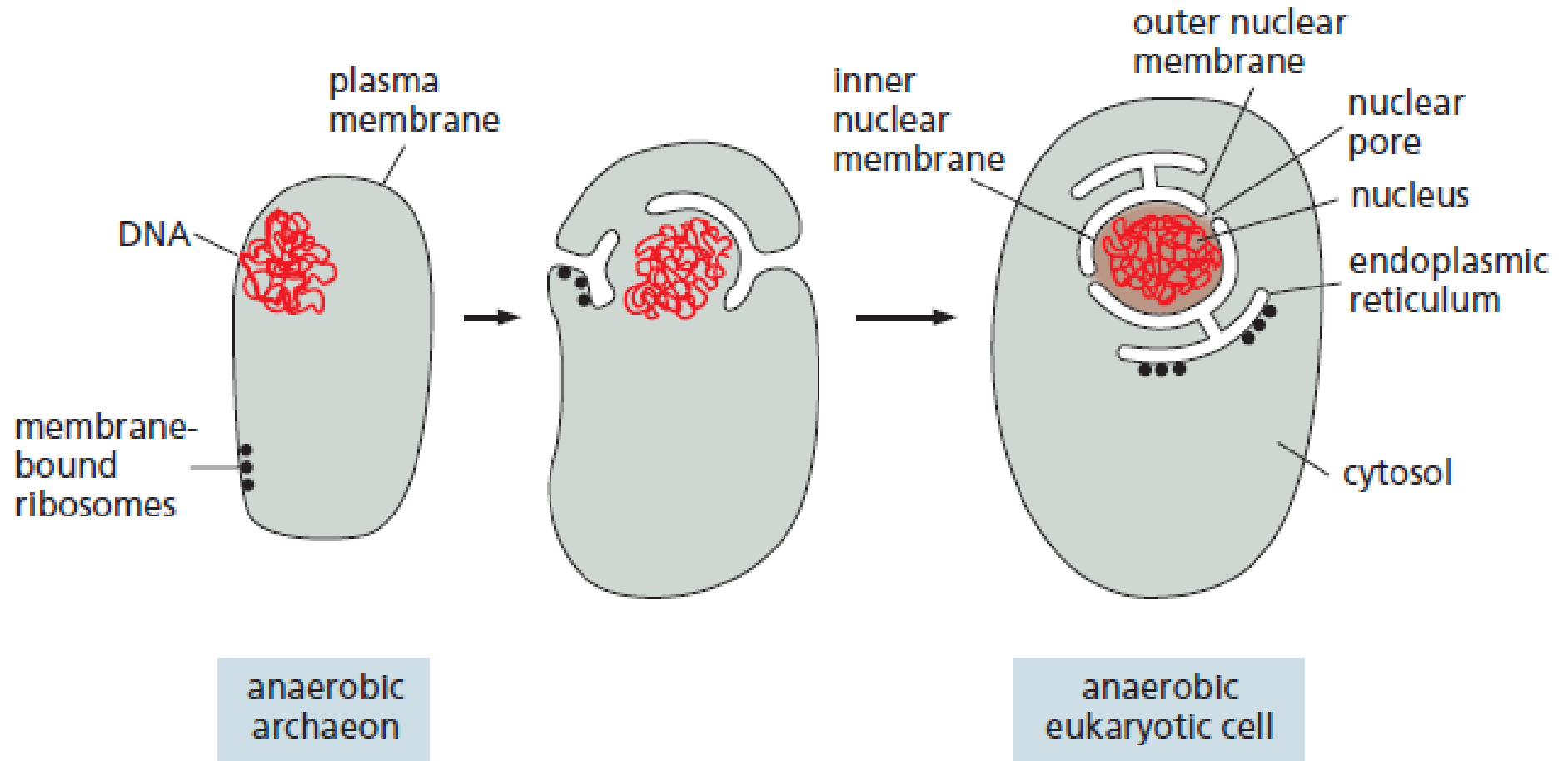
- Fold,
 - Quality check and ER stress
 - Assembly into multi-subunit complexes
- Modified
 - Disulfide bond formation
 - Glycosylation (addition of sugars)
 - Addition of glycolipids to some proteins that are destined for the plasma membrane.





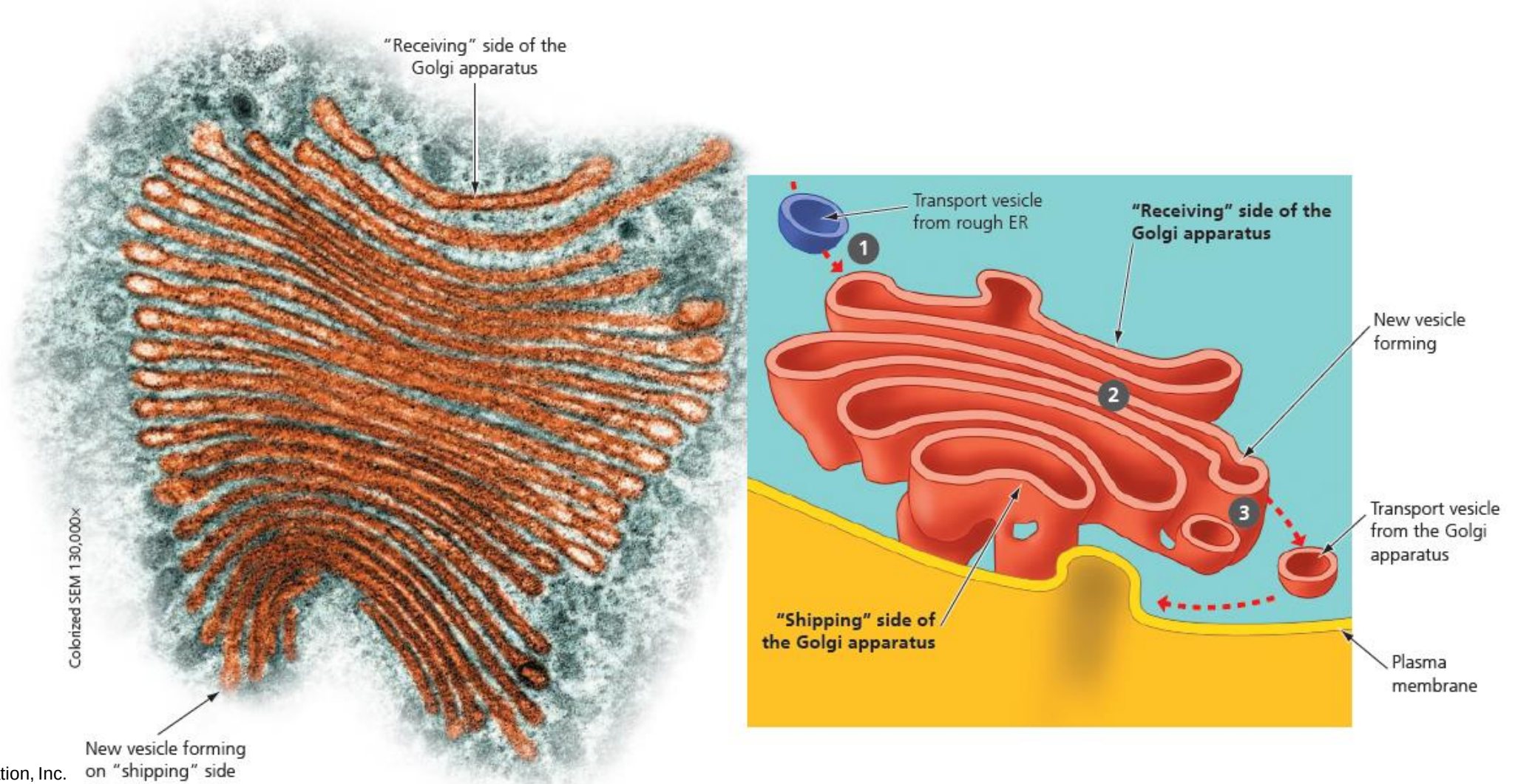
The Endoplasmic Reticulum: Smooth ER

- Lacks surface ribosomes
- Produces lipids, including steroids
 - Cells in ovaries or testes that produce the steroid sex hormones are enriched with smooth ER
- In liver cells detoxify circulating drugs
 - Tolerance to a drug



The Golgi Apparatus

Receives, Refines, Stores, and Distributes chemical products





Maxim Gorky, *Stormy Petrel*

The Nobel Prize in Physiology or Medicine 1906
Camillo Golgi, Santiago Ramón y Cajal



Lysosomes

- A membrane-enclosed sac of **hydrolytic** enzymes found in animal cells.
 - Digestive functions (food vacuoles)
 - Destroy harmful bacteria
 - Engulf and digest parts of another organelle
 - Sculpt tissues during embryonic development
- Most plant cells do not contain lysosomes.

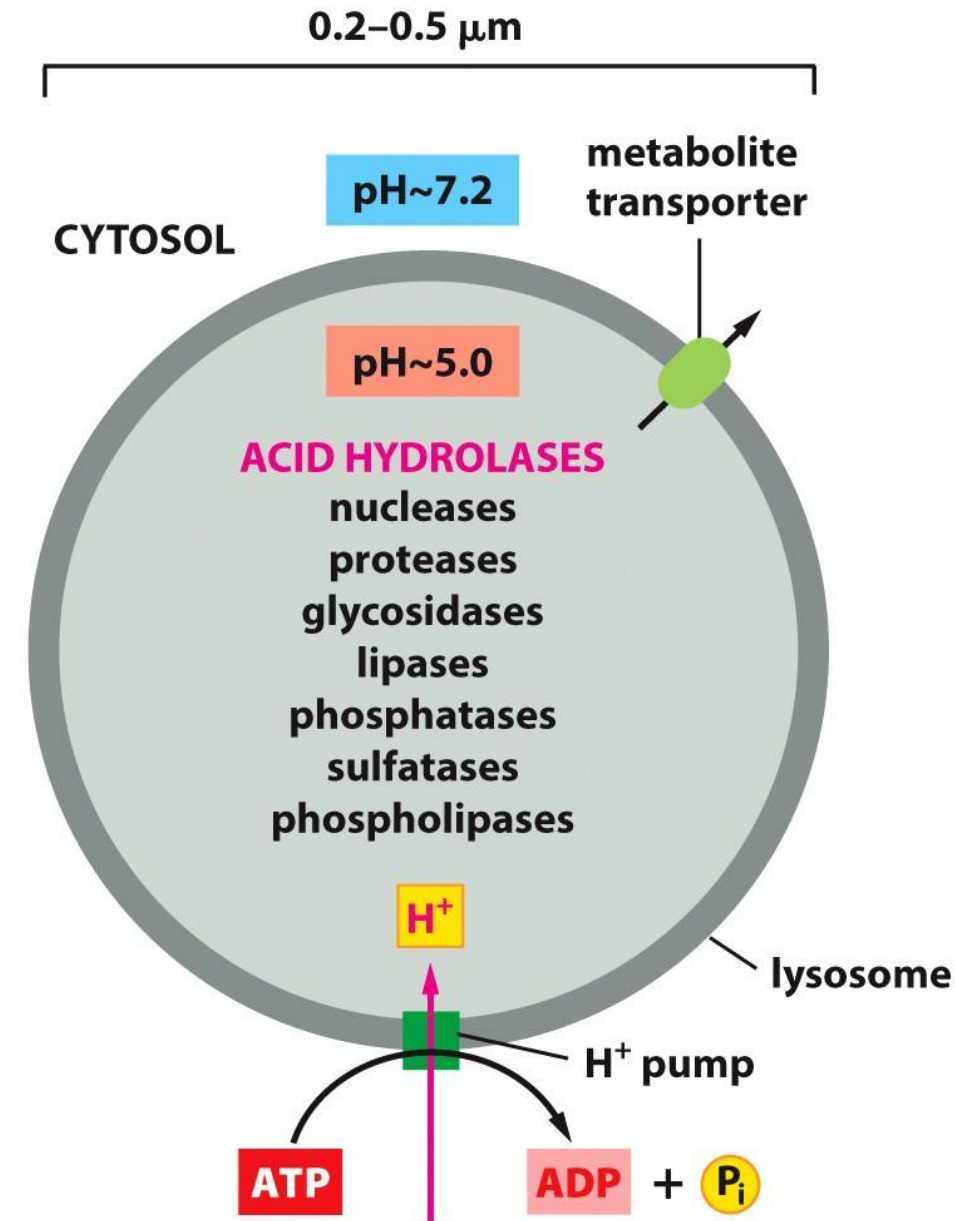
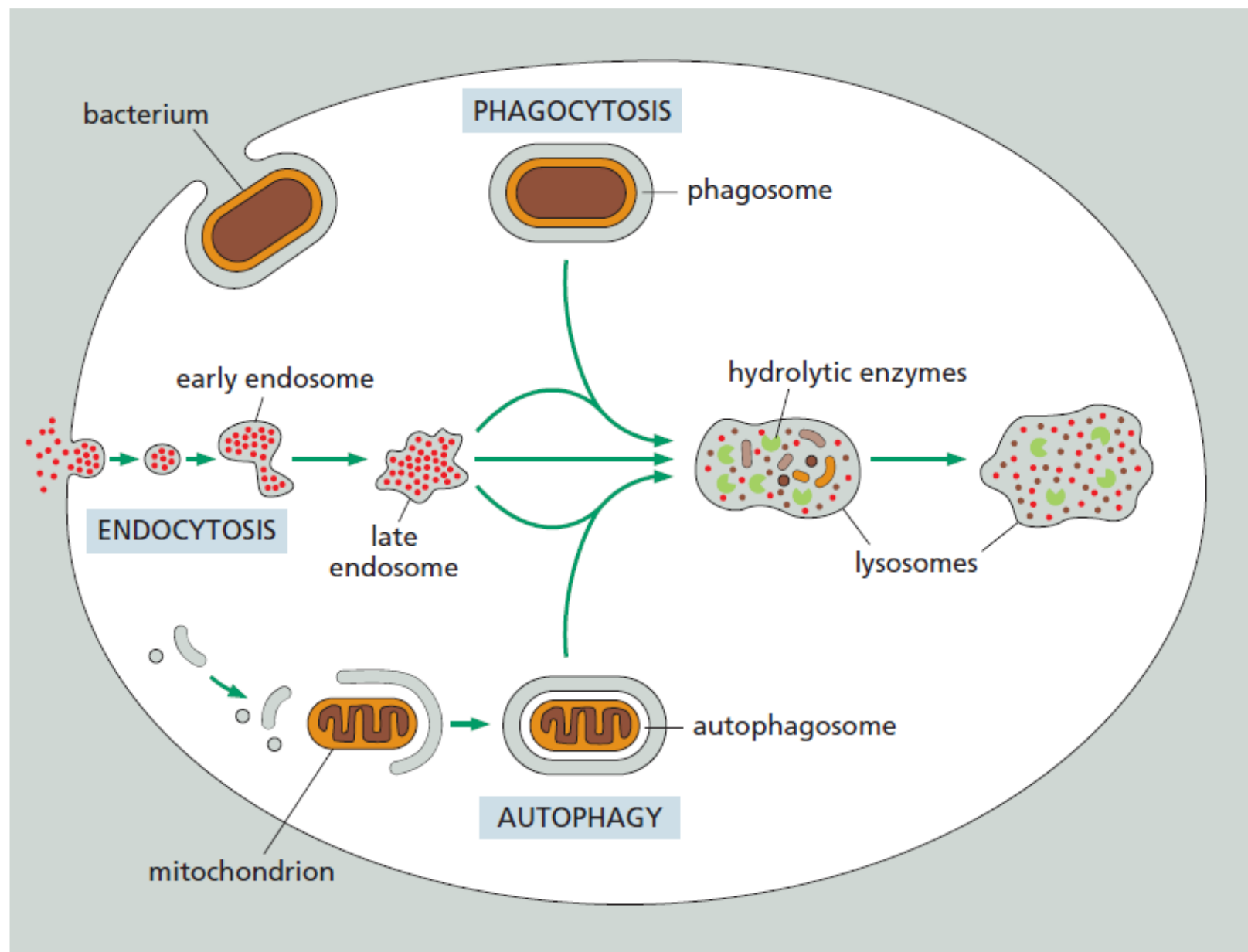
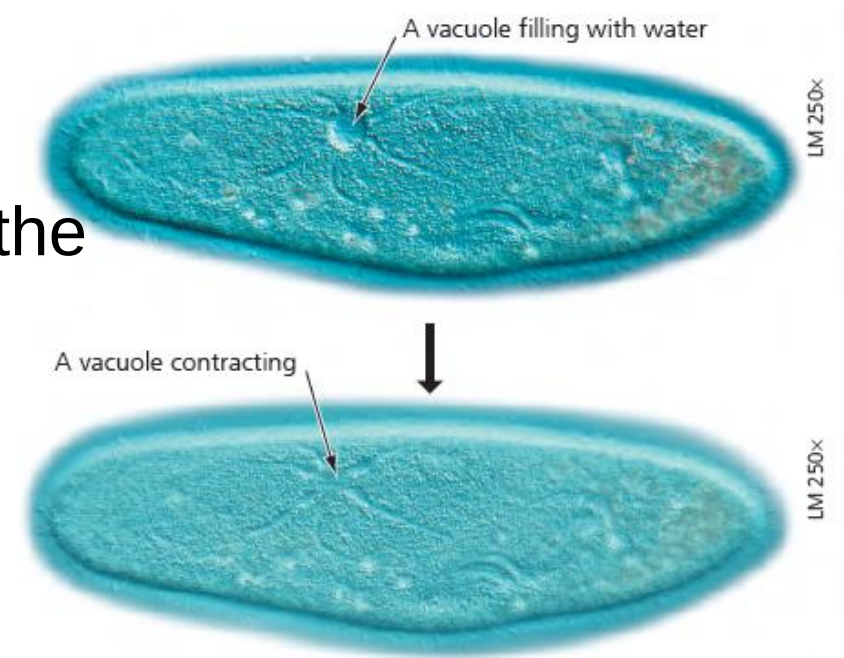


Figure 15–37 Materials destined for degradation in lysosomes follow different pathways to the lysosome. Each pathway leads to the intracellular digestion of materials derived from a different source. Early endosomes, phagosomes, and autophagosomes can fuse with either lysosomes or late endosomes, both of which contain acid-dependent hydrolytic enzymes. Where the membrane fragments that form the autophagosome originate is still actively investigated.



Vacuoles

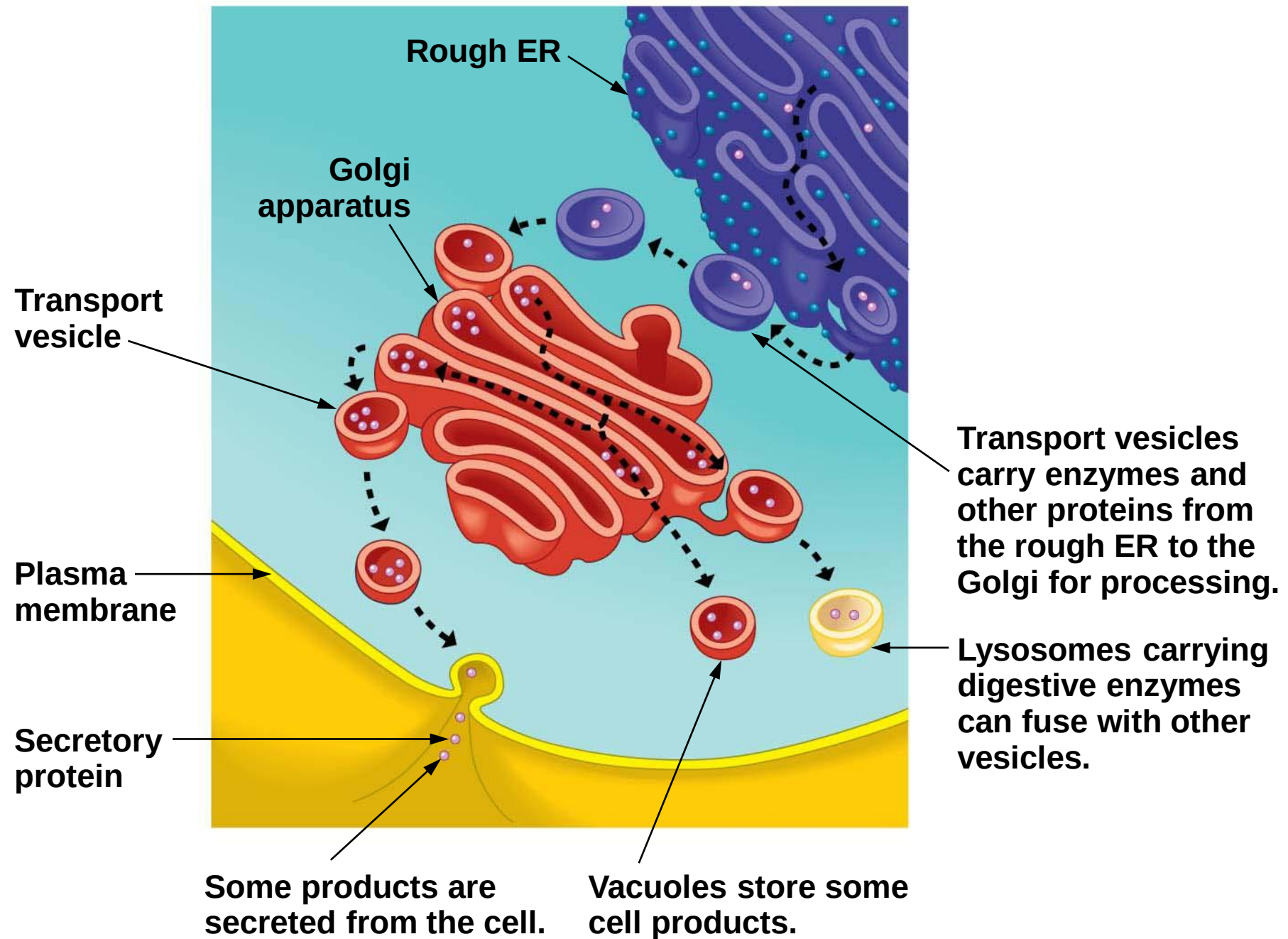
- large sacs made of membrane that bud off from the ER or Golgi apparatus.
- food vacuoles
- contractile vacuoles (伸缩泡)
- Central vacuole in plant cell
 - account for more than half the volume of a mature plant cell
 - store organic nutrients
 - absorb water
 - contain pigments (色素) that attract pollinating insects (传粉昆虫) or poisons (caffeine) that protect against plant-eating animals (herbivores).



(a) **Contractile vacuole in *Paramecium*.** A contractile vacuole fills with water and then contracts to pump the water out of the cell.



(b) **Central vacuole in a plant cell.** The central vacuole (colorized blue in this micrograph) is often the largest organelle in a mature plant cell.



Learning Objectives 3

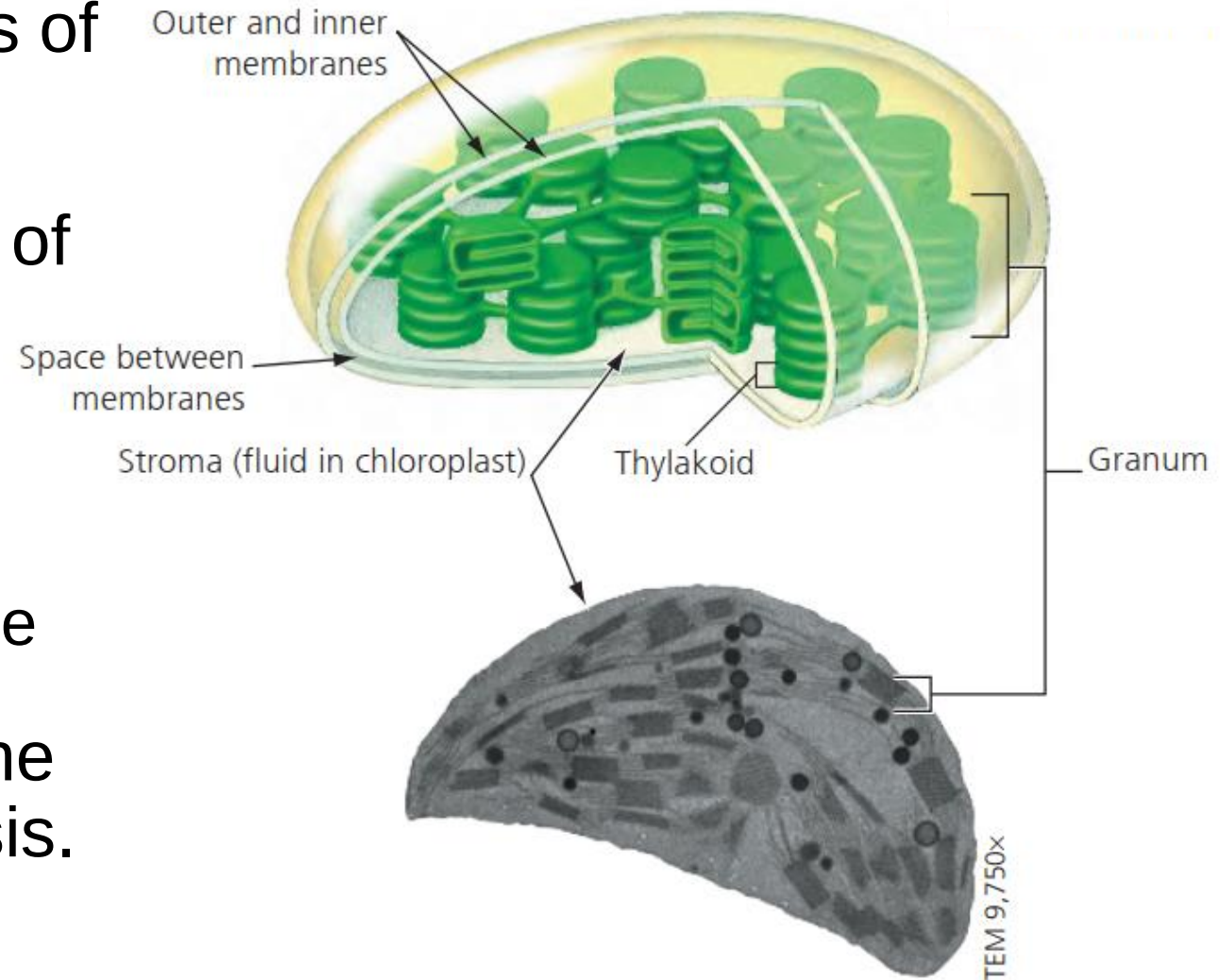
- Compare the structure and function of chloroplasts and mitochondria and explain what structures in chloroplasts and mitochondria result in adaptive advantages.
- Describe the evidence that suggests that mitochondria and chloroplasts evolved by endosymbiosis.
- Describe the functions of the cytoskeleton and compare the structures and functions of cilia and flagella.

Energy Transformations: Chloroplasts and Mitochondria

- A cell converts energy obtained from the environment to forms that the cell can use directly.
- Chloroplasts and mitochondria act as cellular power stations.
- Animal cells have mitochondria but not chloroplasts.
- ~~• So, plant cells have chloroplasts but not mitochondria.~~
- Plant cells have both.

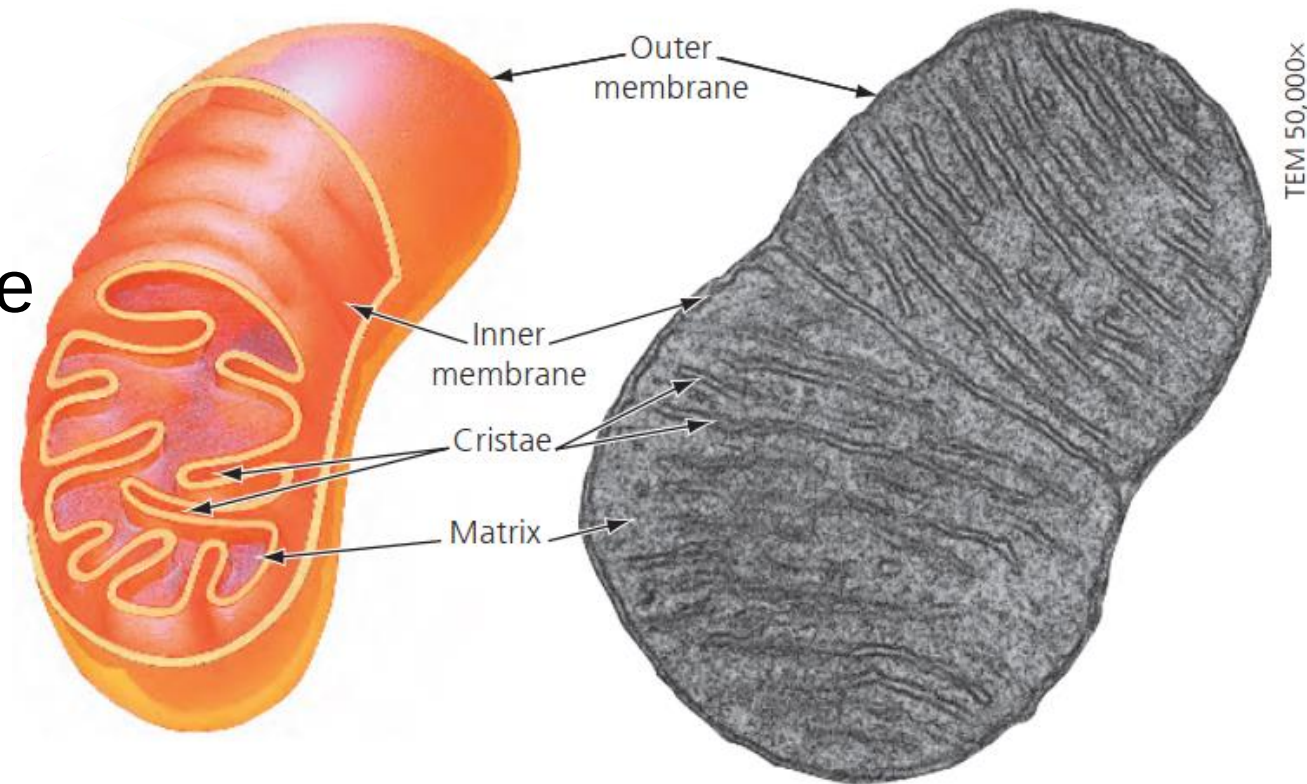
Chloroplasts, the Organelles That Perform Photosynthesis

- Unique to the photosynthetic cells of plants and algae
- Photosynthesis is the conversion of light energy from the sun to
 - The chemical energy of sugar
 - Other organic molecules
 - Occur on the thylakoid membrane
- Most of the living world runs on the energy provided by photosynthesis.



Mitochondria, Site of Cellular Respiration (细胞呼吸)

- In almost all eukaryotic cells
- Produce ATP from the energy of food molecules.
- Cells use molecules of ATP as the direct energy source for most of their work.
- Cristae, infoldings of inner membrane, create greater area

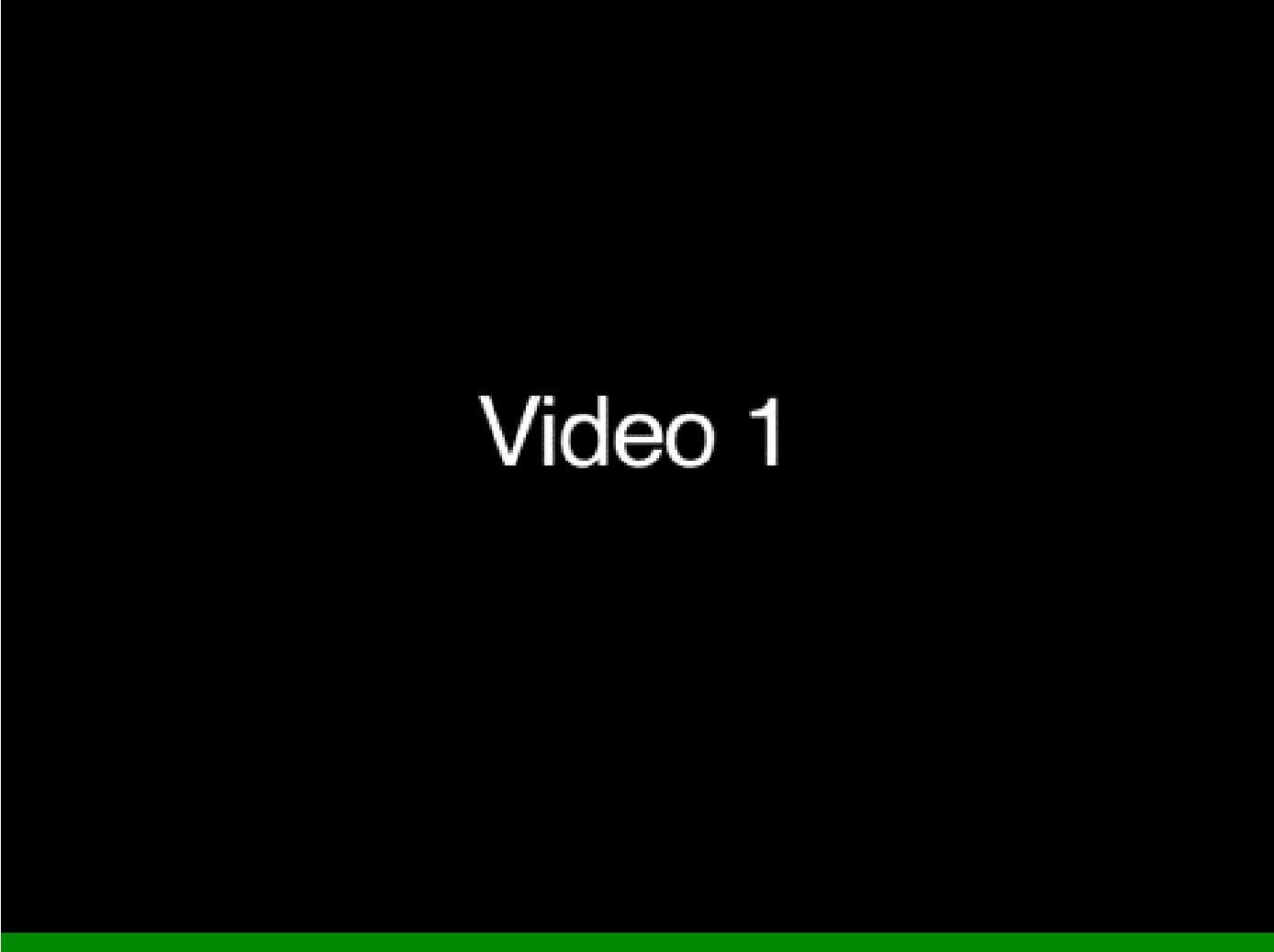


Chloroplast and Mitochondrion

- Contain their **own DNA** that encodes some of their **own proteins** made by their **own ribosomes**.
 - a single circular DNA chromosome
 - *The mitochondrial genome contains 37 genes that encode 13 proteins, 22 tRNAs, and 2 rRNAs.*
- Grow and **pinch** in two, reproducing themselves.
- Evolved from ancient free-living prokaryotes that established residence within other, larger host prokaryotes.
 - This phenomenon, where one species lives inside a host species, is a special type of endosymbiosis (内共生关系).

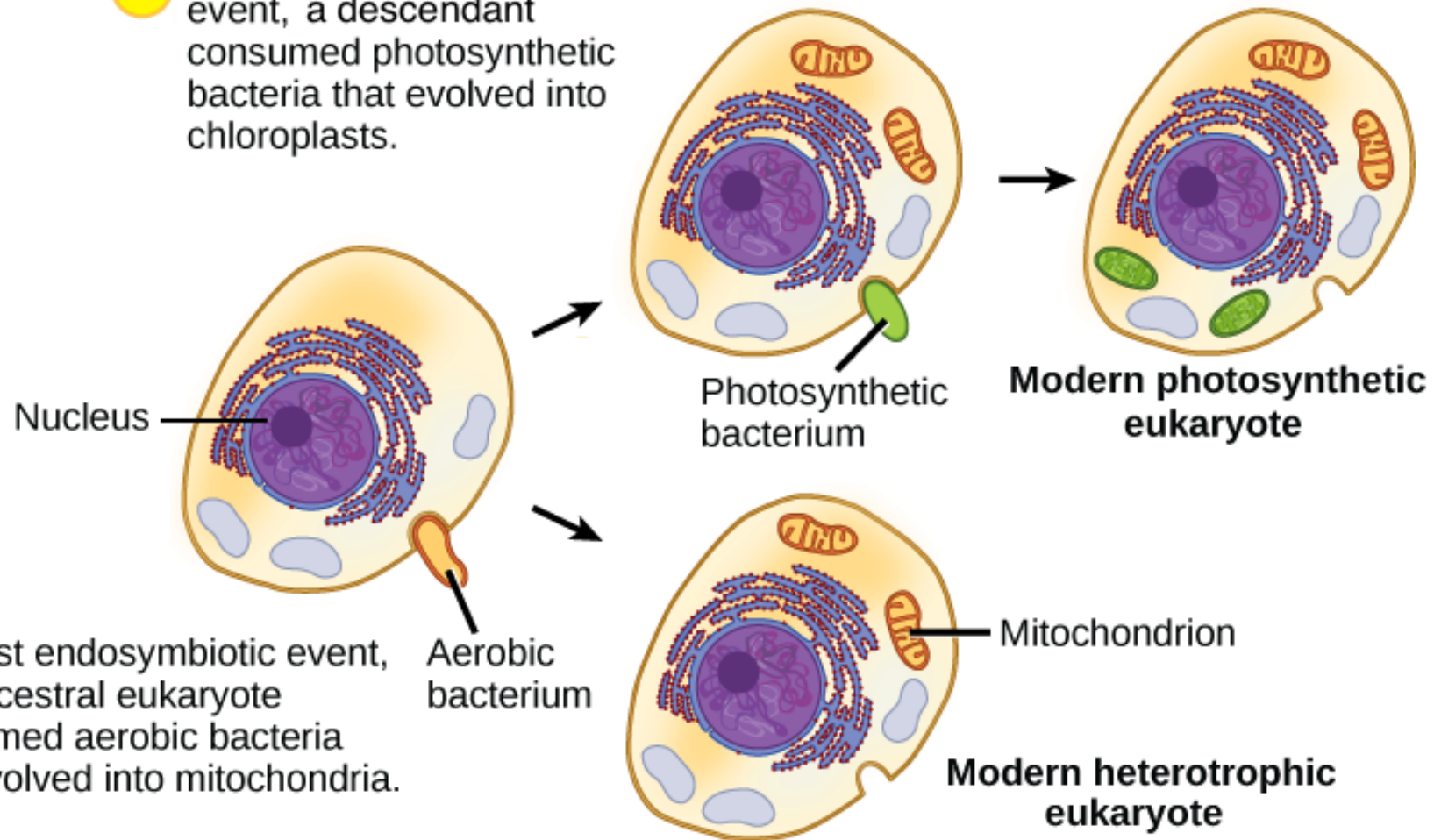
Structure and Dynamics of Mitochondria

Contrary to their illustration as **ovoid structures** in most textbooks, mitochondria are actually dynamic organelles that **fuse and fission** to form constantly changing tubular networks in most eukaryotic cells.



Video 1

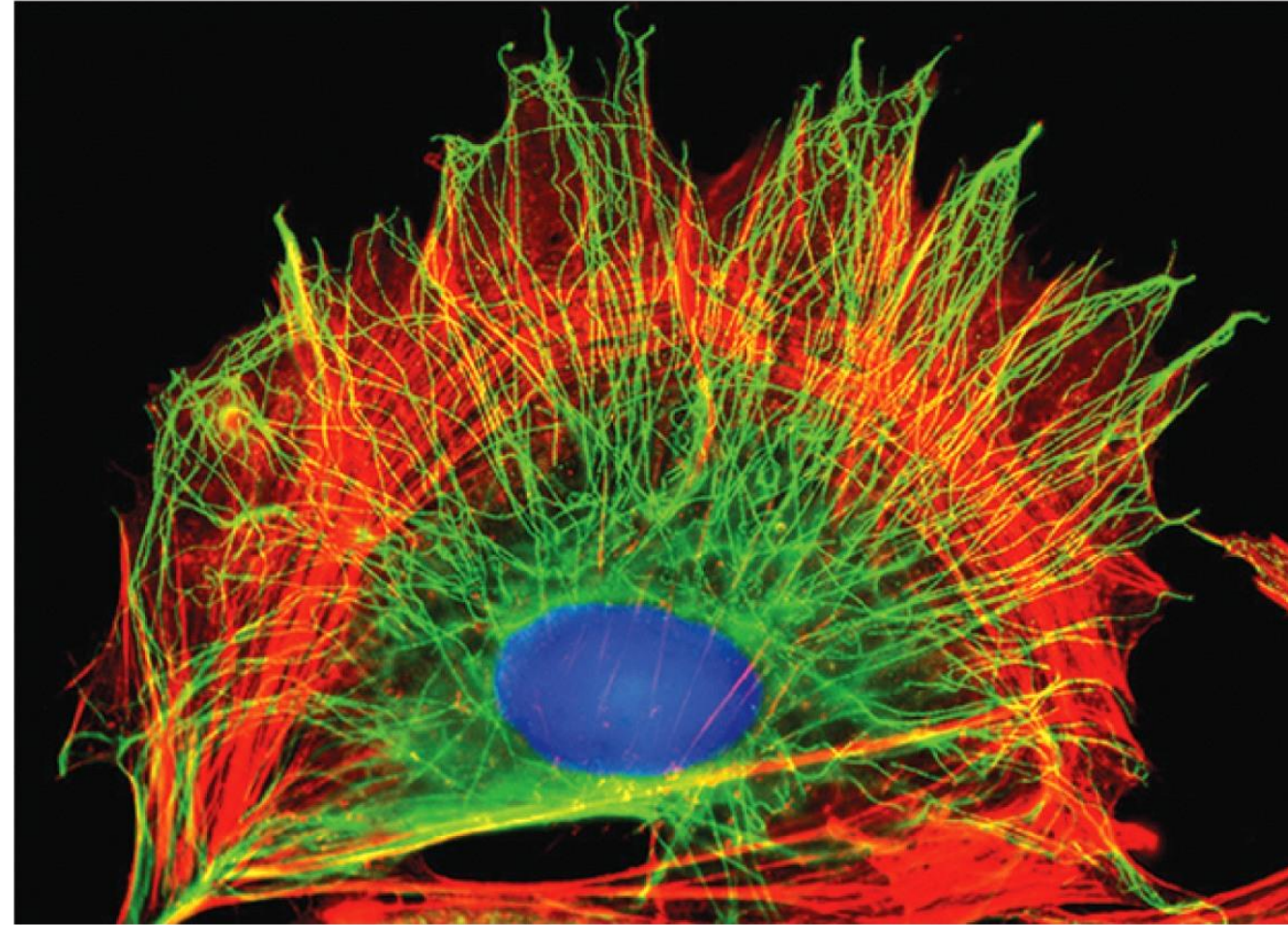
2 In a second endosymbiotic event, a descendant consumed photosynthetic bacteria that evolved into chloroplasts.



1 In a first endosymbiotic event, the ancestral eukaryote consumed aerobic bacteria that evolved into mitochondria.

The Cytoskeleton: Cell Shape and Movement

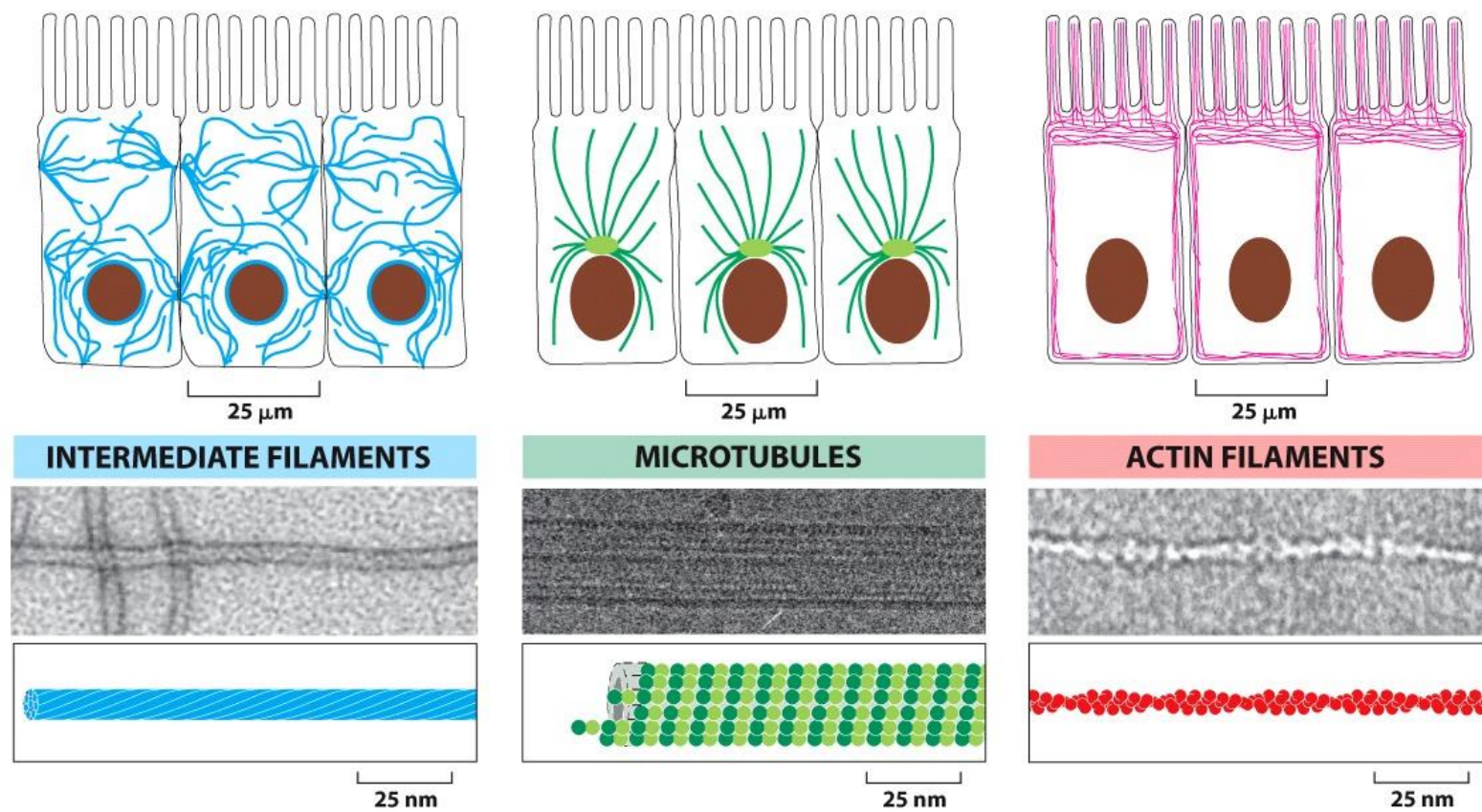
- A network of fibers extending throughout the cytoplasm
- Serves as both “skeleton” and “muscles” for the cell
 - Shape
 - Organize internal components
 - Move



10 μm

Three Types

- Microtubules (微管)
- Intermediate filaments (中间纤维)
- Microfilaments (actin filaments)



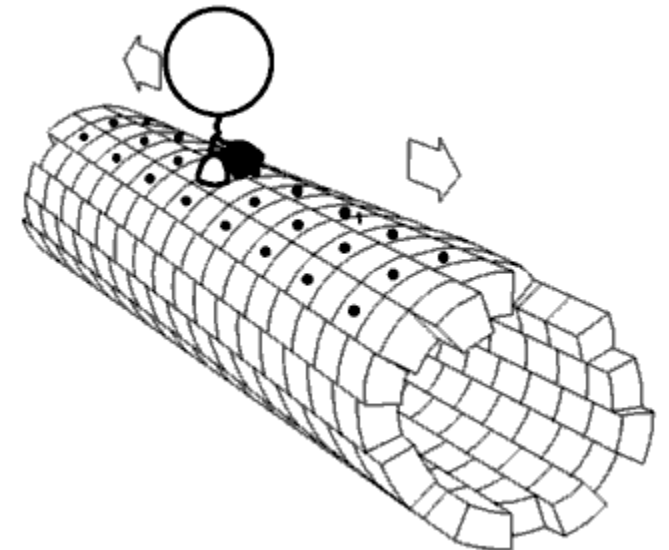
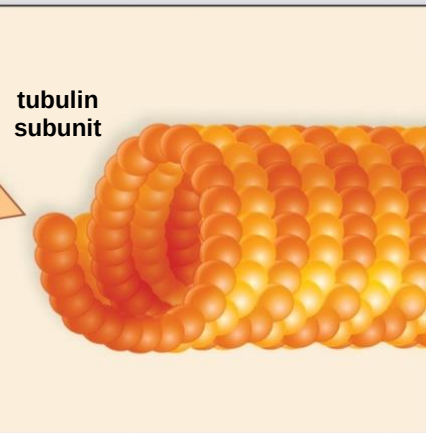
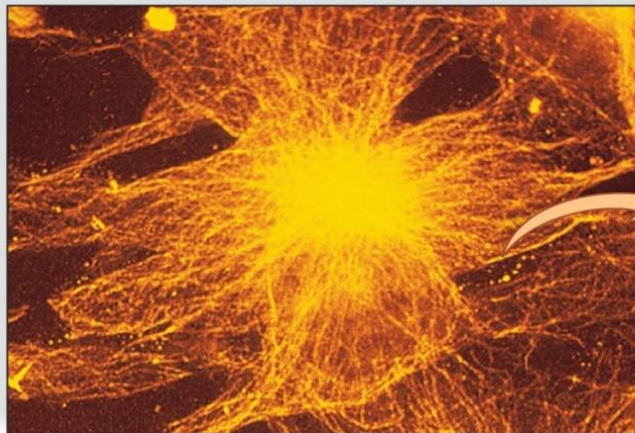
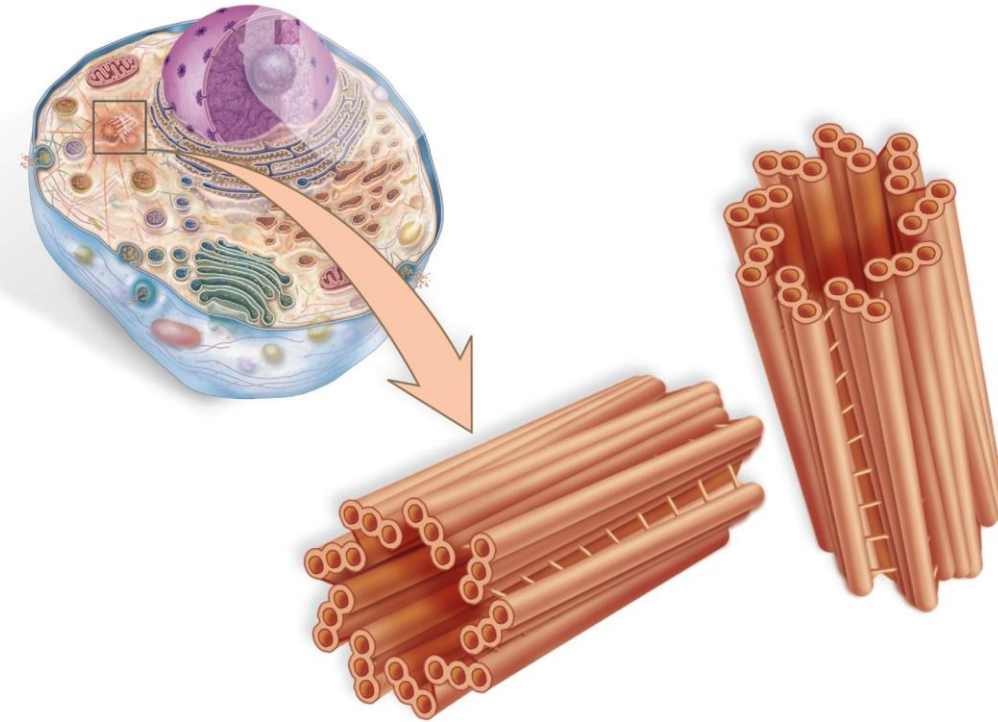
Intermediate filaments are ropelike fibers with a diameter of about 10 nm; they are made of fibrous intermediate filament proteins. One type of intermediate filament forms a meshwork called the nuclear lamina just beneath the inner nuclear membrane. Other types extend across the cytoplasm, giving cells mechanical strength and distributing the mechanical stresses in an epithelial tissue by spanning the cytoplasm from one cell-cell junction to another. Intermediate filaments are very flexible and have great tensile strength. They deform under stress but do not rupture. (Micrograph courtesy of Roy Quinlan.)

Microtubules are hollow cylinders made of the protein tubulin. They are long and straight and typically have one end attached to a single microtubule-organizing center called a *centrosome*. With an outer diameter of 25 nm, microtubules are more rigid than actin filaments or intermediate filaments, and they rupture when stretched. (Micrograph courtesy of Richard Wade.)

Actin filaments (also known as *microfilaments*) are helical polymers of the protein actin. They are flexible structures, with a diameter of about 7 nm, that are organized into a variety of linear bundles, two-dimensional networks, and three-dimensional gels. Although actin filaments are dispersed throughout the cell, they are most highly concentrated in the *cortex*, the layer of cytoplasm just beneath the plasma membrane. (Micrograph courtesy of Roger Craig.)

Microtubules

- Hollow cylinders made of α - and β - tubulin dimer
- Under control of Microtubule Organizing Center (MTOC)
 - In animal
 - Centrosome
 - Basal bodies



Microtubule Dynamics *in vivo*

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Rearrangement

- Quickly dismantled in one part of the cell
- Re-formed in a new location
 - provide rigidity in a new location
 - change the shape of the cell
 - or even cause the whole cell or some of its parts to move

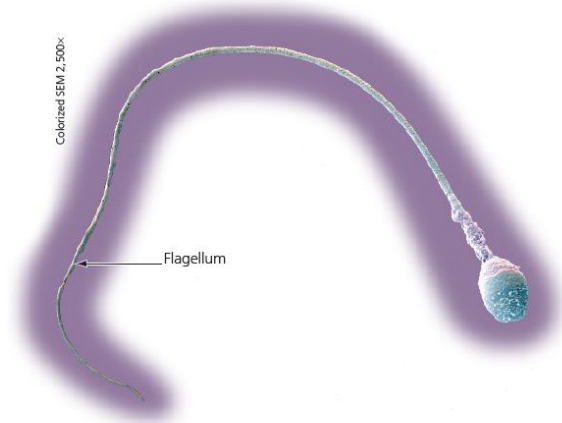
Crawling Amoeba

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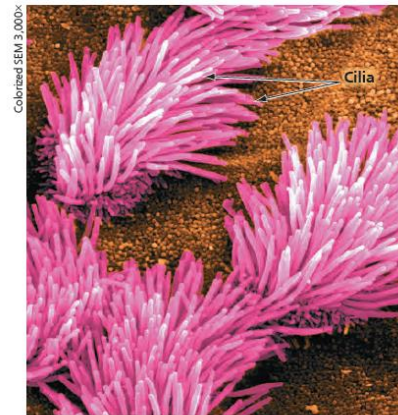
Cilia (纤毛) and Flagella by Microtubules

- In some eukaryotic cells, microtubules are organized by **basal bodies** into flagella and cilia, extensions from a cell that aid in movement.
- Eukaryotic flagella propel cells through an undulating, whiplike motion.
- They often occur singly, such as in human sperm cells (精子), but may also appear in groups on the outer surface of protists.

▼ Figure 4.21 Examples of flagella and cilia.



(a) **Flagellum of a human sperm cell.** A eukaryotic flagellum undulates in a whiplike motion, driving a cell such as this sperm cell through its fluid environment.



(b) **Cilia lining the respiratory tract.** The cilia lining your respiratory tract sweep mucus with trapped debris out of your lungs. This helps keep your airway clear and prevents infections.

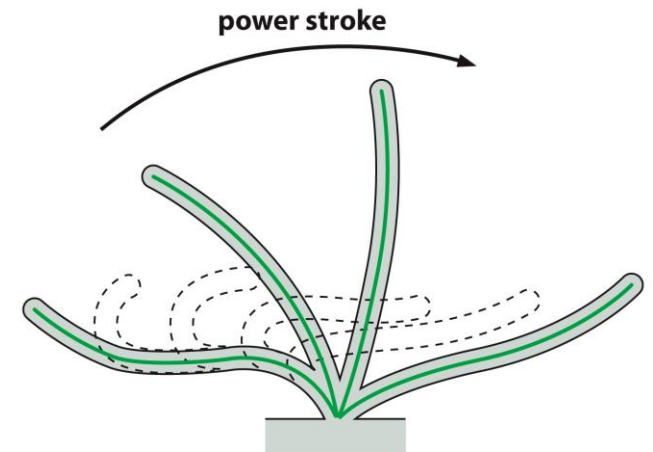
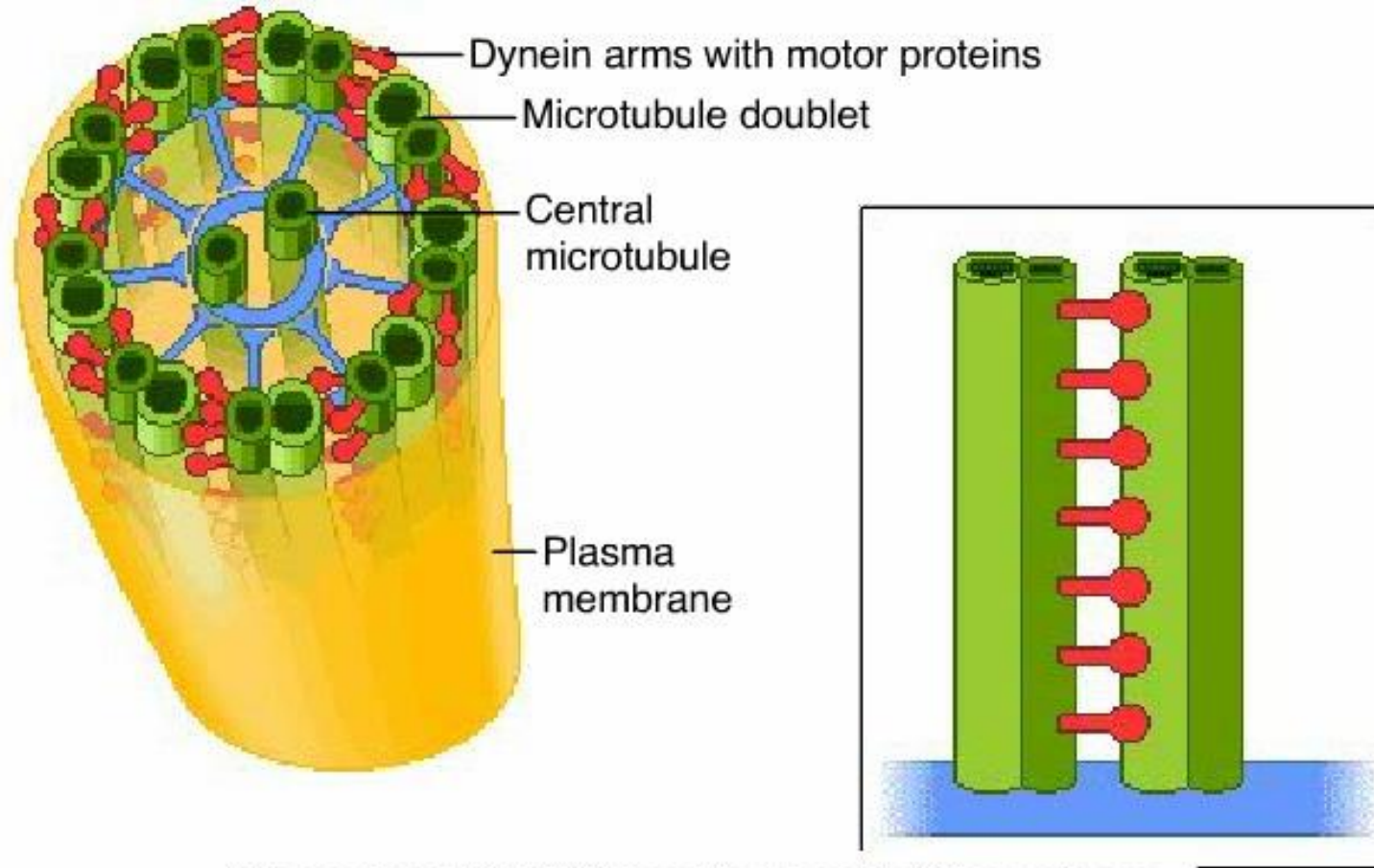


Figure 17-24. Essential Cell Biology, 4th ed. (© Garland Science 2014)

Cilia and Flagella



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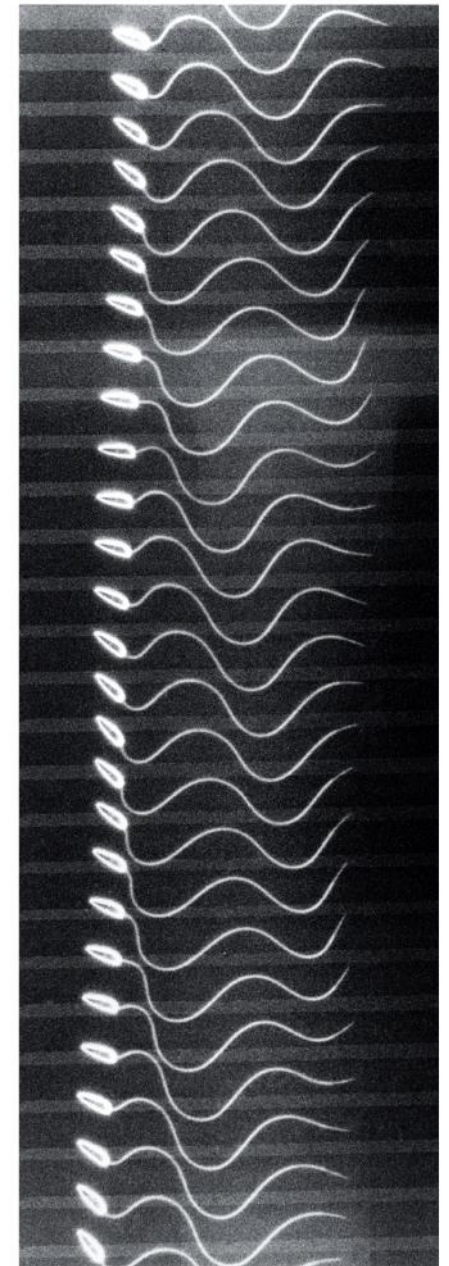
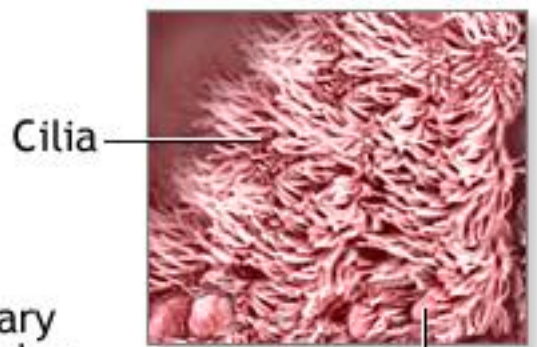


Figure 17-25 Essential Cell Biology, 4th ed. (© Garland Science 2014)



Hair-like projections called cilia line the primary bronchus to remove microbes and debris from the interior of the lungs



Cilia

Primary bronchus

Goblet cell

ADAM.

**Movement
of mucus
to pharynx**



**Ciliated columnar
epithelial cell**

Mucous cell

Stem cell

Mucous gland

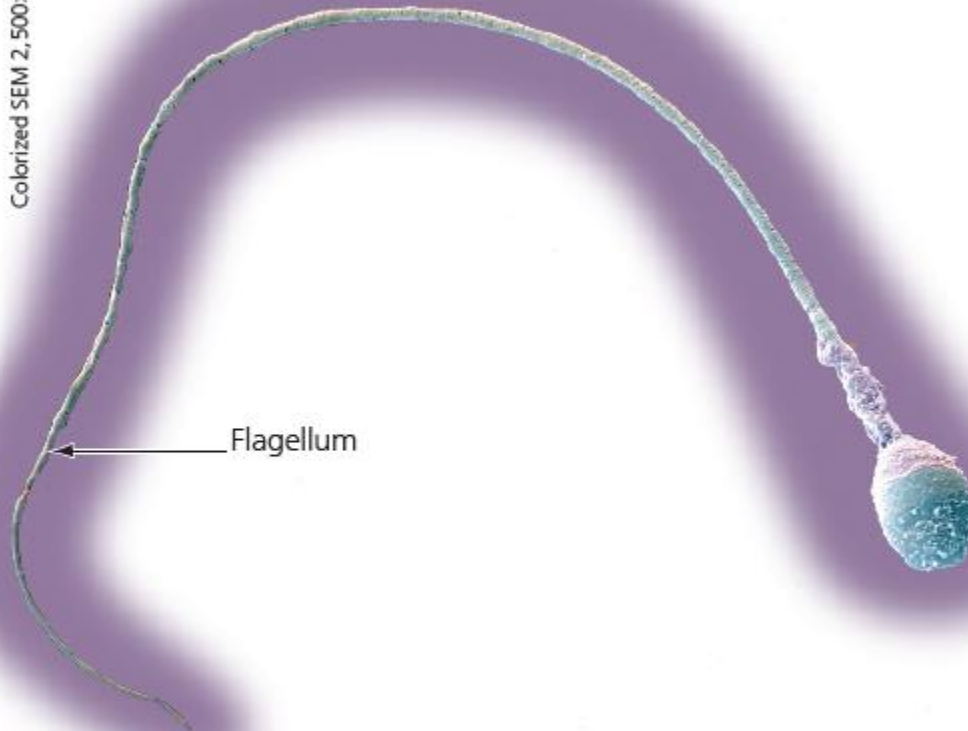
Mucus

b A diagrammatic view of the respiratory epithelium of the trachea, showing the direction of mucus transport inferior to the pharynx.

Why do some men with a type of hereditary sterility (遗传性不育) also suffer from respiratory problems?

▼ **Figure 4.21** Examples of flagella and cilia.

Colorized SEM 2,500×



(a) Flagellum of a human sperm cell. A eukaryotic flagellum undulates in a whiplike motion, driving a cell such as this sperm cell through its fluid environment.

Colorized SEM 3,000×



(b) Cilia lining the respiratory tract. The cilia lining your respiratory tract sweep mucus with trapped debris out of your lungs. This helps keep your airway clear and prevents infections.

Why do some men with a type of hereditary sterility (遗传性不育) also suffer from respiratory problems?

Because of a defect in the structure of their flagella and cilia, the sperm of men afflicted with this disorder cannot swim normally within the female reproductive tract to fertilize an egg (causing sterility) their cilia do not sweep mucus out of their lungs (causing recurrent respiratory infections).